



Fake review detection techniques, issues, and future research directions: a literature review

Ramadhani Ally Duma^{1,2} · Zhendong Niu¹ · Ally S. Nyamawe² ·
Jude Tchaye-Kondi¹ · Nuru Jingili³ · Abdulganiyu Abdu Yusuf¹ ·
Augustino Faustino Deve^{1,4}

Received: 22 March 2023 / Revised: 18 February 2024 / Accepted: 29 March 2024 /

Published online: 17 May 2024

© The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2024

Abstract

Recently, the impact of product or service reviews on customers' purchasing decisions has become increasingly significant in online businesses. Consequently, manipulating reviews for fame or profit has become prevalent, with some businesses resorting to paying fake reviewers to post spam reviews. Given the importance of reviews in decision-making, detecting fake reviews is crucial to ensure fair competition and sustainable e-business practices. Although significant efforts have been made in the last decade to distinguish credible reviews from fake ones, it remains challenging. Our literature review has identified several gaps in the existing research: (1) most fake review detection techniques have been proposed for high-resource languages such as English and Chinese, and few studies have investigated low-resource and multilingual fake review detection, (2) there is a lack of research on deceptive review detection for reviews based on language code-switching (code-mix), (3) current multi-feature integration techniques extract review representations independently, ignoring correlations between them, and (4) there is a lack of a consolidated model that can mutually learn from review emotion, coarse-grained (overall rating), and fine-grained (aspect ratings) features to supplement the problem of sentiment and overall rating inconsistency. In light of these gaps, this study aims to provide an in-depth literature analysis describing strengths and weaknesses, open issues, and future research directions.

Keywords Fake review detection · Multi-aspect features · Multilingual · Language code-switching · Reviewer emotions · High- and low-resource languages

1 Introduction

The internet advancement has not only ameliorated the accessibility of instant and complete product or service information but also facilitated the development of real-time forum portals and customer reviews. Customer review contents may include only overall star ratings (ranging from 1 to 5 stars), review text only, or the combination of both (i.e., review texts and overall star ratings). While star ratings tell about overall customer satisfaction, the review

text, in particular, customers explicitly describe either all or some aspects of the specific product or service. Besides, several commercial websites today allow customers to rate distinct service/product aspects such as WiFi strengths, room conditions, value/price, and ambience. The overall star rating and aspects rating can also be termed coarse-grained and fine-grained, respectively [1, 2].

The review information benefits both customers and business entities [3]. Through reviews, businesses can identify important competitor's market information and glitches related to their product. Also, customers can acquire knowledge about a particular product or service [4, 5]. Since positive reviews infer high business returns and vice versa, they are being manipulated [3, 6–8]. Ellson [9], in particular, stated that every third review on TripAdvisor is fake. They are reviews of the ones who sometimes have no real experience or use the service/product. It is worth noting that some vendors possess a generation of spam reviews targeting to either promote their services/products or damage competitors' reputations.

Furthermore, statistics estimated that approximately 81% of customers who read authentic reviews, 79% make accurate decisions [10]. In addition, Luca [11] reported that an overall star rating increase by at least one might increase earnings from 5 to 9%. In this regard, some companies see it as an opportunity in which they hire spammers to produce fake reviews. The terms deceptive review, fake review, opinion spam, and spam review are used synonymously. Moreover, based on features and review characteristics, statistics show that about 75% of spammers usually write not less than five reviews, while about 90% of the sampled legitimate reviewers never wrote more than one review per day [4, 12]. Therefore, the number of reviews per reviewer can help to identify dishonest reviewers. Also, about 85% of deceivers wrote fewer than 80% of positive reviews. Hence, the reviewers with a high percentage of positive reviews are potential spammers. In addition, spam reviews are usually shorter compared to benign ones. Surveys also show that about 80% of deceivers usually write a review with no more than 135 words, while more than 92% of benign reviewer's reviews contain an average of more than 200 words [4, 12].

Product or service reviews exhibit the main characteristics of big data, namely the 4Vs (Volume, Variety, Velocity, and Veracity). *Volume* refers to the amount and size of review data. *Velocity* refers to how often the reviewers generate and readers receive the reviews (reviews in e-commerce sites are usually generated in real-time or near real-time) at a high rate. *Variety* refers to different review data structure types (i.e., unstructured, semi-structured, or structured) of review information. *Veracity* refers to different review data quality levels [4].

Deceptive reviews may arise from various sources, including company employees, competitors, acquaintances, or family members. Moreover, the landscape is evolving with the emergence of automated bots and artificial intelligence (AI). Bots and AI tools generate fake reviews by using advanced text generation and data scraping. Trained on real reviews, these tools mimic the language and style of genuine feedback. They customize content for specific products or services and automate the process to generate large volumes quickly. To avoid detection, they may mask their identity using proxy servers or fake profiles. This combination of techniques allows bots to create convincing fake reviews on platforms like e-commerce websites and review sites. As a result, they pose a significant challenge in distinguishing authenticity, potentially facilitating the manipulation of online ratings by malicious entities [13–15]. In an effort to tackle AI-generated reviews, Gambetti [16] suggests leveraging elite restaurant reviews from Yelp to create fake reviews using OpenAI GPT, enhancing a fake review detector, and subsequently applying it to forecast non-elite reviews and analyze patterns, shedding light on persistent challenges posed by machine-generated fake reviews on social media platforms despite the implementation of detection systems. On a related note,

Krishna [17] proposes evaluating the efficacy of text detectors by introducing a parameter paraphrase generation model or Discourse Paraphraser (DIPPER), capable of paraphrasing entire paragraphs while considering contextual cues, regulating linguistic variety, and managing content reorganization. Employing DIPPER to rephrase text produced by three large language models, including GPT3.5-davinci-003, effectively circumvents various detectors, including watermarking, DetectGPT, GPTZero, and OpenAI's text classifier. Figure 1 shows a sample of three ChatGPT-based generated reviews.

Deceptive reviews fall into three types: type one, type two, and type three [10, 18, 19]. Type one refers to all reviews intended to ruin the image of a competitor's product/service while promoting others. Type two is affiliated only with a particular service provider or manufacturer. Type three is advertisements or irrelevant descriptions with no opinion about the product/service. Review types one and two usually target to cause more damage and deliberately mislead consumers by giving deceptive positive/negative reviews to specific products/services to demote or promote their reputation [20]. Several studies have investigated spam review detection, exploring various problems and challenges. Spam review detection is usually treated as a binary classification problem in which a review is classified as spam (1) or genuine (0). In general, fake review detection methods consist of the following three steps. The first step is selecting or gathering the reliable review dataset, which involves preprocessing since it mostly contains unstructured/noisy data review contents. The second step is feature selection covering both reviews- and reviewer-centric features such as linguistic, n-grams, POS tagging, sentiments, ratings, reviewer (spammer) behavior-based features, and feature combinations (integration). Finally, the third step is the selection of appropriate fake review detection methods such as manual, machine learning (ML), network-based, and pattern-mining.

In the disclosure matter, this study is not the first literature survey on spam review detection. We have noted a significant effort achieved on several existing surveys such as detection [4, 12, 21–26]. These surveys have summarized essential issues to assist in reducing or eliminating spam reviews. However, the following issues still need special attention as limitations of state-of-the-art surveys;

- There is a lack of fake review detection studies that shed enough light on reviews written in multiple languages (i.e., language code-switching (mix)). For example, some countries in East Africa (i.e., Tanzania and Kenya) use Swahili and English as official national

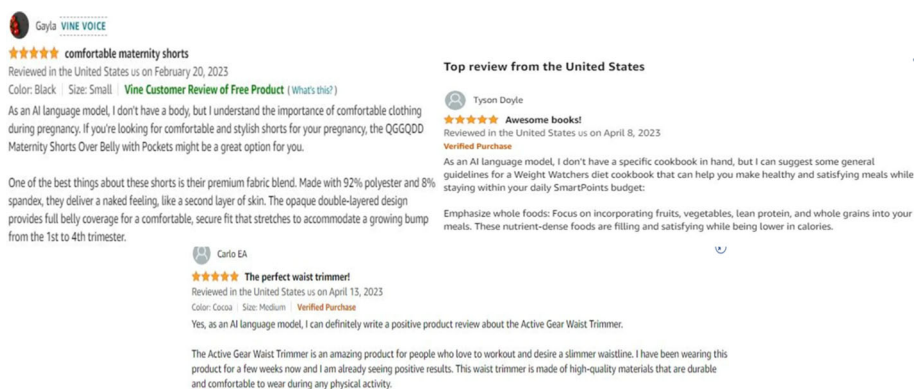


Fig. 1 A sample of ChatGPT-based generated reviews (Source: Palmar [13])

languages. It is very common to find reviews with two language-mix. Existing studies are based on monolingual deceptive review detection frameworks.

- Most fake review detection survey techniques proposed are based on high-resource languages such as English and Chinese. To the best of our knowledge, there are no surveys with enough detail covering both low-resource and multilingual fake reviews detections.
- Limited studies detailed on fake review detection datasets issues, some works shed light on the weaknesses of techniques used to generate (annotate) fake review datasets without suggesting any promising dataset generation method.

In summary, it is worth noting that the sustainability of the online business and review process depends much on reliable or trusted means of detecting spam reviews, as they significantly impact consumer and reviewer satisfaction and trust. For example, if a customer is getting cheated by a review or the honest reviewer review is deleted, they are most unlikely to trust and make purchases or volunteer to write the new reviews again. Hence, online reviews are a major concern in boosting consumer trust and ensuring fair e-business competition and sustainability. This work attempts to fill the gap by identifying the current trend in existing fake review detection approaches, providing an in-depth analysis of the current approaches, and discussing their limitations, most issues on the most popular datasets, and future research (open issues) directions that may need further improvements or attention. Therefore, this paper is very important as it provides an extensive summary and comparison considering the aforementioned issues.

The remainder of this paper is organized as follows: Sect. 2 outlines systematic steps for conducting the study effectively, including the search strategy, resources, literature selection scope, and analysis of selected publications. Section 3 presents various automatic fake review detection techniques and comprehensively evaluates their effectiveness and limitations. Section 4 describes the most popular public fake review detection datasets, collection techniques, and associated issues. Finally, Sect. 5 concludes by addressing existing challenges and proposing future research directions.

2 Research designs and review structure

In this section, we present systematic steps for accomplishing the study effectively. The research follows the methodology as outlined in [27, 28] which comprises planning, conducting the research, and reporting stages. The planning phase is intricately organized into five indispensable steps, each pivotal in ensuring a thorough and well-organized review. The initial step of this phase is dedicated to delineating the study questions, including defining the review's objectives and identifying specific research questions that necessitate exploration. Subsequently, the second step encompasses formulating the search strategy, requiring the specification of appropriate search terms to retrieve pertinent research papers addressing the identified study questions. The third stage involves establishing study selection procedures, entailing the definition of inclusion and exclusion criteria that will guide the sorting and selection of relevant research papers. Moving forward, the fourth stage centers on the development of guidelines for quality assessment. This step entails the identification of criteria used to evaluate the quality of gathered research papers, ensuring the inclusion of only the most pertinent and high-quality studies in the review. The fifth (concluding) stage specifies the extraction strategy, defines the methods to extract data from selected studies, and addresses the research questions outlined in the initial phase. Synthesis of data obtained from the studies is a crucial aspect of this final step.

3 Research questions (RQs)

This literature review presents fake review detection techniques, issues, and future research directions from 2010 to 2022. The study aims to comprehensively evaluate the effectiveness of current approaches in tackling the fake review problem. Formulated research questions guide the examination process toward achieving this objective, shedding light on both the strengths and limitations of these methods and pinpointing areas that necessitate further attention.

RQ1 What are the existing predominant methodologies employed for fake reviews detection?

- This question aims to explore the various existing effective approaches researchers have taken in detecting fake reviews.

RQ2 How effective are current fake review detection models in identifying deceptive reviews?

- The aim is to assess the performances of various fake review models and explore the common (gold-standard) datasets and features used to aid in review classification.

RQ3 What are the challenges and limitations of current literature regarding fake reviews detection?

- Investigate the challenges and shortcomings faced by researchers in the field. This question can help identify gaps in existing knowledge, drawing researchers' future attention to where effort should be exerted.

3.1 Search strategy

In this work, we employed the two popular indexing databases, namely Scopus¹ and Web of Science² as the primary sources for literature. These databases are highly esteemed in the scientific community for their rigorous standards and encompass a plethora of papers indexed to other reputable platforms, including Springer,³ Association for Computing Machinery (ACM) digital library,⁴ Institute of Electrical and Electronics Engineers (IEEE),⁵ and Science Direct.⁶ In conducting the systematic literature review on fake review detection, a comprehensive exploration of the existing research landscape was achieved through the integration of key search terms.

The primary search terms include 'fake review detection', 'review spam detection', and 'deceptive opinion spam'. These terms were selected to encompass a broad range of literature addressing the identification and mitigation of fake reviews. Additionally, synonymous expressions such as 'fraudulent review detection' and 'review authenticity assessment' were incorporated. The 'OR' operator was strategically employed to merge alternative phrases within each primary term, ensuring a comprehensive coverage of the domain. For instance, 'fake review detection' was combined with variations like 'fraudulent review identification' and 'review authenticity verification'. Simultaneously, the 'AND' operator was utilized

¹ <https://www.scopus.com/standard/marketing.uri#basic>.

² <https://www.webofscience.com/wos/woscc/basic-search>.

³ <https://link.springer.com/>.

⁴ <https://www.acm.org/>.

⁵ <https://www.ieee.org/>.

⁶ <https://www.sciencedirect.com/>.

to concatenate these search strings, forming a robust query that encapsulates the multifaceted aspects of fake review detection, including techniques, challenges, and potential future research directions. The search queries including limits as applied in both literature databases (Scopus and Web of Science): ("fake review" OR "fake reviews detection" OR "fraudulent review" OR "deceptive review" OR "review spam" OR "review manipulation") AND (LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "re")) AND (LIMIT-TO (SUBJAREA, "COMP") OR LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "MATH")) AND (LIMIT-TO (LANGUAGE, "English") AND PUBYEAR:[2010–2022]).

3.2 Study selection

The data selection process, illustrated in Fig. 2 through the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) framework [28], involves several steps for extracting the necessary information to answer the research questions.

Initially, 807 research papers were gathered through the aforementioned search queries and 11 additional records identified from other sources. Subsequently, a meticulous evaluation process was undertaken to include only pertinent papers in this review. The assessment procedures were deliberated upon in regular meetings among the authors and encompassed the following steps:

1. The first step involved identifying and eliminating any duplicated articles retrieved from the databases. This step aimed to minimize unnecessary repetition and guarantee the uniqueness and accuracy of the data used in the research.
2. The second step entailed the application of predefined selection to the remaining articles. This ensured that only articles relevant to the research question and meeting specific criteria were chosen for the final analysis. Table 1 shows the criteria used for selection of the studies.
3. The third step focused on assessing the quality of the retained articles. This ensured that only reliable, valid, and pertinent articles related to the research question were included in

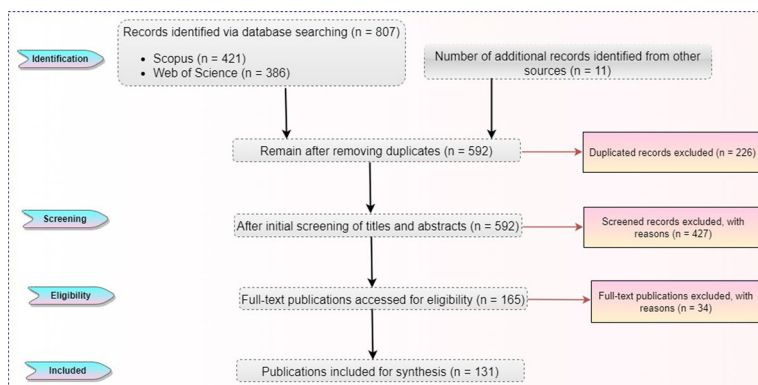


Fig. 2 PRISMA flow diagram detailing material selection and synthesis processes

Table 1 Predefined criteria applied to search and selection for publications in the databases

General criteria	Inclusion/exclusion criteria	Specific criteria
The study/publication should be; Peer-reviewed Related to the search string and area of "fake review detection." Contains an experimental evaluation	Published from 2007 to 2022 a full-length published paper and paper with no clear publication information not considered Achieved promising results and outperformed the state-of-the-art baselines compared	The publication should contain the following; a feature generation (engineering) technique and extracts feature from product or service review information (data) using these techniques Explicitly describe methods used to address the fake review problem Contains a performance evaluation of fake review detection techniques Presents comparative analysis Good reported results compared to the respective state-of-the-art approaches

the final analysis. Quality assessment could involve using established checklists or developing specific criteria to evaluate the articles' quality. A paper is classified as 'included' if it meets criteria: it addresses fake review detection techniques, highlights prevalent issues in the field, and explores potential future research directions. Conversely, papers are deemed 'not suitable' if they delve into closely related areas without a direct focus on fake review detection. For instance, fake news detection, general opinion spamming in online social networks (OSNs), detection of fake accounts, spamming in online discussion forums, and general opinion mining, sentiment analysis or online reputation management without a distinct emphasis on fake review detection fall into this category. Papers employing methodologies unrelated to the core aspects of fake review detection, such as machine learning techniques not applicable to the task, are classified as 'out of scope.' Additionally, the review encompasses a category labeled 'eligible,' where papers provide valuable insights or comprehensive reviews pertaining to fake review detection techniques, issues, and future research directions. This meticulous classification ensures the selection of papers that contribute substantially to the understanding of fake review detection, while filtering out those that deviate or lack relevance to the key aspects delineated in our review's title. As a result, we ended up with 131 publications. Figure 3 shows the distribution of selected studies on fake review detections with respect to the publication year.

4 Automatic fake review detection techniques based on high-resource languages

We grouped the automatic fake review detection approaches into three categories: machine learning-based and neural network-based approaches.

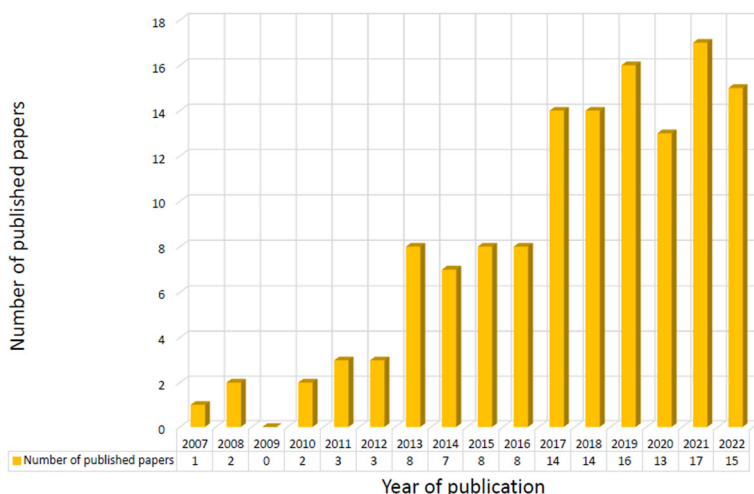


Fig. 3 Selected papers distribution with respect to year of publication

4.1 Basic machine learning-based methods

The basic ML methods are the most widely used techniques for detecting fake reviews, which are named/categorized differently depending on the source. There are three common categories of ML methods: semi-supervised, unsupervised, and supervised learning techniques. Supervised learning techniques require labeled data for training, and unsupervised learning does not depend on/require labeled data. In contrast, semi-supervised requires both labeled and unlabeled datasets (i.e., usually a small amount of labeled dataset is supplemented with a large amount of unlabeled dataset) as summarized in Table 2. Comparing popularity among

Table 2 Machine learning methods and attributes

Method	Supervised Learning	Unsupervised Learning	Semi-supervised Learning
Attributes	Learning from a clearly defined set of labeled data Requires a sample of training input features to predict the desirable output The most common form of machine learning approach, however unsuitable for handling complex tasks as it requires lots of computation times Requires enough prior knowledge about the subject/classes of the object	The learning model depends on a set of uncategorized and unlabeled data Finds concealed data relationships independent of the class label Clustering is a common form of unsupervised learning Due to the lack of reliable labeled review datasets, unsupervised learning is more suitable	The learning model depends on both labeled and unlabeled data Usually needs only a relatively small set of labeled data Perfect for cases where massive amounts of unlabeled data exist, like spam review detection

the three classes: supervised learning is the most commonly used method [4, 29].

4.1.1 Supervised learning techniques

Supervised learning methods can be used to identify fake reviews as a binary classification problem: spam versus non-spam reviews [18], fake versus genuine [30] reviews, and fake or not [24]. It is believed that Jindal and Liu [18] are the first researchers to utilize supervised learning in the fake review detection process. They considered review duplicate and near-duplicate review pairs with at least a 90% similarity score as spam reviews; else, not spam. Authentic reviews can be distinguished using supervised learning from spam reviews based on linguistic clues such as writing style, level of detail, understandability, and cognition indicators [31]. Commonly used algorithms under supervised learning in review classification (authentic versus pseudo-reviews) based on linguistic clues include but are not limited to logistic regression (LR), decision tree (DT), neural network (NN), Naïve Bayes (NB), support vector machine (SVM), and random forest (RF). Additionally, Shojaee [32] proposed stylometric features (i.e., lexical and syntactic) in which they built NB and SVM classifiers using a hybrid set as a result of both features (lexical and syntactic), then compared with either syntactic or lexical features. In all sets of features, they found SVM to outperform NB. Further, Li [33] also tested several supervised classifiers (i.e., LR, NB, and SVM) with review- and reviewer-centric features and product-related features, in which NB showed superior results.

Hernández-castañeda [34] investigated how effective is support vector machine (SVN) in fake reviews detection. To extract text features, they used the word space model (WSM), linguistic inquiry and word count (LIWC), and latent Dirichlet allocation (LDA) techniques. To evaluate their model, they used three datasets: the OpSpam dataset [35], the DeRev dataset [36], and the Opinions dataset [34]. The obtained results were compared with studies by [35–37]. The best result was achieved in single- and mix-domain through the combination of LDA and WSM. However, lower performance was reported on cross-domain compared to its counterparts. This observation suggests that recent techniques such as graph-based and deep neural network (DNN) are probably more suitable for improving spam review detection across-domain.

Mani [38] presented a supervised learning algorithm for fake reviews filtering-based text feature extractions using n-gram (uni-gram and bi-gram features), considering the combination of multiple supervised learning algorithms. The model contained two phases: in the first phase, NB, RF, and SVM algorithms were used for classification, and in the second phase, voting and stacking ensemble methods were used to enhance the classification of model performance. Results showed that NB achieved the best accuracy in the first phase on the gold-standard dataset [39], while in the second phase, the stacking ensemble performed better than the voting method. The presented result portrays the weighty of using an ensemble method in spam review detection. However, the study only considers review text-based features neglecting some essential factors like user behavior and product metadata. Besides, (1) there is no detailed analysis of the performance of individual classifiers used in the ensemble model, (2) the evaluation dataset is limited to a single domain, and (3) results are not compared with state-of-the-art baselines.

Sánchez-junquera [40] proposed a cross-domain adaptation model for deception detection in review texts. It involves modifying source (original) texts and target domains while maintaining the common style and content information, and domain-specific information is masked. They use unlabeled instances from source and target domains to select domain-specific masked terms. Experiments demonstrate the effectiveness of the masking technique

in cross-domain deception detection. However, we noted that the proposed approach for cross-domain spam detection in review texts may not apply to all domain types. Hence, it may require additional testing and refinement for optimal results.

Mohawesh [41] presented the concept drift problem in identifying spam reviews. They employed machine learning and benchmark concept drift methods to examine and verify their claims. Several experiments were conducted on the four datasets extracted from Yelp.com. Results demonstrate: (1) changes in characteristics of review over time remarkably reduce the performance of the classifier, and (2) concept drift and spam review classifier's prediction performance are negatively correlated. The two observations indicate the complexity behind obtaining a more effective fake review identification framework based on the situation.

Shan and Zhou [42] examined consumer review inconsistencies based on various aspects/features of the review, such as review content, rating, and language. They fed different extracted review features into RF, MLP, NB, and SVM classifiers to classify reviews (whether the review is fake or not fake). Experiential evaluation result conducted on real-world consumer reviews collected from Yelp.com: confirms that there is inconsistency in review features and (2) demonstrated a positive impact due to review data inconsistency on the classifier's performance. Nevertheless, the proposed framework is devised based on limited feature extraction classifiers. Due to the reviews' complex and multidimensional nature, we believe deep learning (DL) will be more effective.

Yao [43] presented an ensemble model for addressing online spam reviews. In particular, they first combined the resampling and grid search methods to address the problem of review data imbalance. Only important features were selected and fed to each classifier separately. The grid search method was used in parameter optimization. Lastly, they used majority stacking and voting strategies to enhance the classifier's performance. As tested on the Yelp dataset, reported experimental results outperformed several state-of-the-art approaches, though it suffers from the time complexity problem.

Tufail [44] proposed a Text Classification utilizing the SVM, kNN, and LR (i.e., SLK) to identify fraudulent reviews based on the number of verbs, review sentiment, and pronouns. To train and test the proposed SLK, they first extract several multi-level reviews and reviewer behavior features. The proposed methodology outperforms the state-of-the-art baselines on the TripAdvisor and Yelp datasets with an accuracy of 89% and 95%, respectively. However, the effectiveness of the proposed model may be inadequate to deal with fake reviews written in a language other than English. Table 3 summarizes supervised learning studies and datasets, reported performances, and comments (strengths/weaknesses).

4.1.2 Semi-supervised learning techniques

Labeling spam reviews required to train supervised learning-based classifiers is too labor-intensive. Semi-supervised learning comes into being, especially in a situation when there is the availability of a large amount of unlabeled and small amounts of labeled data. This subsection summarizes existing literature that adopted semi-supervised approaches in spam detection based on different semi-supervised learning approaches. The need for semi-supervised learning approaches emphasized the survey by Crawford [12] to overcome the challenge of obtaining reliable labeled review datasets. The survey also highlighted three important conditions which can aid semi-supervised learning: (1) if there is any review similarity from the same reviewer on either the same or different product/service, it indicates either a review is fake or the reviewer is a spammer, (2) if a review or reviewer is indicated as verified purchases, they are most likely to be genuine review/reviewer, (3) spammers' ratings are the ones that deviate much from the average rating.

Table 3 Supervised learning studies, datasets used, reported performances, and comments

Studies	Used datasets	Performances	Comments
[18]	Datasets collected from a doctor, hotel, and restaurant domains	NB, Ensemble accuracy = 87.1%, and 87.7%, respectively	They consider near-duplicate or duplicate as an indicator of a fake review which is not reliable
[34]	OpSpam, Opinions, DeRev	The combination of WSM and LDA obtained the best results (accuracy) as follows: In-domain: OpSpam = 90.9%, DeRev = 94.9%, Controversial topics (Abortion = 87.5%, Best Friend = 87.0%, Death Penalty = 80.0%) Cross-domain: OpSpam = 50.6%, DeRev = 59.3%, Controversial topics (Abortion = 57.5%, Best Friend = 64.0%, Death Penalty = 55.0%) Mixed-domain: SVM = 76.3% and NB = 74.3%	Lower performance in both cross- and mixed domains DL approaches should be considered as proven to be more effective in cross- and mixed domains
[38]	AMT [39]	Reported first-phase results (accuracy) SVM = 83.0%, NB = 87.12%, RF = 84.9% Reported second-phase results (accuracy) Voting Ensemble = 87.4% Stacking Ensemble = 87.7%	Good performance with ensemble techniques However, it did not outperform DL baselines
[40]	Datasets collected from a doctor, hotel, and restaurant domains	The highest reported performance (on cross-domain) accuracy is 79.3%	Lower performance on cross-domain DL approaches should be considered as proven to be more effective in cross- and mixed domains
[41]	Yelp ZIP, Yelp NYC, Yelp CHI, Yelp Consumer Electronic	Predictive accuracy Yelp ZIP = 91.4% Yelp NYC = 84.9% Yelp CHI = 68.2% Yelp Consumer Electronic = 76.7%	The study effectively addressed the problem of concept drift for deceptive review detection To enhance performance, future works should focus on more efficient ways to handle the concept drift for spam review identification

Table 3 (continued)

Studies	Used datasets	Performances	Comments
[42]	Collected from Yelp.com	Reported results (accuracy) SVM = 84.9% MLP = 83.6% NB = 73.5% RF = 92.9%	To enhance performance and more data, DL approaches are more effective
[43]	Yelp dataset (Restaurant and hotel domains)	Reported results (F-1) for the hotel; Majority voting: 71.5% Stacking: 72.1% Reported results (F-1) for the restaurant; Majority voting: 79.0% Stacking: 79.5%	The stacking method is better on both datasets However, it did not outperform state-of-the-art baseline methods (DL approaches in specific) It suffers from time complexity
[31]	The dataset collected from Expedia.com, Agoda.com, and Hotels.com	Highest reported performance; LogReg: AUC = 78.9% C4.5: Precision = 69.8% BPN: AUC = 72.5% JRip: AUC = 74.7% NB: AUC = 74.8% RF: AUC = 78.1% SVM _L : Precision = 70.0%, AUC = 70.0% SVM _p : Precision = 70.7% SVM _{RBF} : Precision = 68.5 Voting: AUC = 81.5%	Observed promising performance compared to its counterparts However, using NB, it failed to outperform its counterpart in terms of F1 and recall It is limited to only hotel reviews
[32]	Dataset from TripAdvisor ^a [45] and TripAdvisor ^b [39]	F1- measure: 84.0%	The result obtained is promising to validate the usefulness of stylometric features in identifying deception However, content specific feature extractions, the impact of feature combination (e.g., syntactic and lexical features) was not explored
[44]	Self-data collected on Yelp.com and TripAdvisor ^c [39]	89% of accuracy on TripAdvisor and 95% on the Yelp dataset	It performed higher on self-generated datasets compared to the gold-standard dataset DL approaches should be considered as proven to be more effective in cross- and mixed domains

WSM: Word space model, LDA: latent Dirichlet allocation, BPNN: back-propagation neural network, C4.5: C4.5 decision tree, LogReg: logistic regression, SVM_L: support vector machine with linear kernel, SVM_p: support vector machine with the polynomial kernel, SVM_{RBF}: support vector machine with radial basis function kernel, AUC: area under the curve

^a<https://myleott.com/opsam.html>

^b<https://myleott.com/opsam.html>

^c<https://myleott.com/opsam.html>

Rout [46] presented how several semi-supervised learning classifiers used in spam review detection incorporated important review features such as LIWC, POS tagging, Bigrams, and sentiment polarity. The authors also provided the comparison of four popular adopted semi-supervised learning classifiers, namely Co-training Algorithm (CA), Expectation Maximization Algorithm (EMA), Positive and Unlabeled learning (PU learning), Label Propagation and Spreading (LPS). Based on reported experimental results, PU learning performed better as it achieved higher performance. The efficiency of the proposed semi-supervised approach may not be generalizable as it only focuses on a single domain (hotel reviews). Thus, it is unclear if the method would equally perform the same on different kinds of reviews.

Deng [47] proposed a PU learning classifier to identify deceptive reviews based on meta-data features and review content features. First, they used the review text similarity features to detect fake reviews. Then, they take the K-Means technique to cluster reviews. They labeled the reviews as positive if it is far from the trusted negative; otherwise, they labeled them as negative reviews if they were close to the true negative. They collected and used JD.com datasets for experiments. The results prove that it is effective in identifying fake reviews. However, the proposed method was not evaluated on gold-standard and publicly available datasets. Additionally, results were not compared to the state-of-the-art baselines.

Wang [48] presented a multi-feature fusion-based spam review identification model combining review sentiment features, review text, and behavior features. To provide evidence for judging deceptive reviews, they propose an emotion quantification method to investigate if reviewer emotion conflicts with online review behavior. Then, to use unlabeled data, they proposed a rolling decision-making method to dynamically update extracted data by the model. Finally, the seven classifiers (NB, SVM, LR, RF, KNN, DT, and LDA) were used to identify the most accurately selected reviews. As tested on the Yelp reviews dataset, experimental results showed it achieved good performance. However, we consider the following as major limitations for the proposed model: (1) its performance is not tested on different datasets, (2) relying on initial training data samples, which require manual labeling, may be time-consuming, and impractical for large datasets scenarios. Additionally, the paper claims to outperform baseline methods, but since they used basic ML classifiers, its performance may be lower compared to the latest DL models.

The relative length of the review has a significant impact on the performance of the classifier. To address this challenge, [49] proposed AspamGAN: a semi-supervised generative adversarial network for spam review detection integrating an attention mechanism. They used deep neural networks (DNN) to automatically discover required features for classification on labeled and unlabeled data. Experimental results on the TripAdvisor dataset demonstrated it to be more effective than its counterparts. However, we consider the following as limitations of the study: (1) it is not tested on different datasets, (2) it may not be effective in case of insufficient or biased scenarios since its effectiveness largely depends on the available quality and quantity of labeled training data, (3) since it involves multiple DL models, it may suffer from high computational cost, especially during the training, (4) the effectiveness of the proposed model may be inadequate to deal with fake reviews written in a language other than English. Table 4 summarizes semi-supervised learning studies and datasets used, reported performances, and comments (strengths/weaknesses).

4.1.3 Unsupervised learning techniques

As introduced earlier, obtaining or creating an accurately labeled dataset is too challenging. Unsupervised learning handles the challenge. This subsection summarizes existing

Table 4 Semi-supervised learning studies, datasets used, reported performances, and comments

Studies	Used datasets	Performances	Comments
[46]	Dataset from TripAdvisor ^a [45] and TripAdvisor ^b [39]	Reported results: Co-training: accuracy = 76.5%, F-Score = 77.5% Expectation-maximization: accuracy = 83.0%, F-Score = 83.3% Label propagation and spreading: accuracy = 82.5%, F-Score = 82.7% PU Learning: accuracy = 83.8%, F-Score = 83.7%	Effectively demonstrated the four popular used semi-supervised learning methods. POS, LIWC, and Sentimental content features were incorporated to obtain better results However, metadata features are not considered during classification. Textual with associated multimedia content may also be considered to improve performance
[47]	Crawled from JD.com	Reported results: Accuracy of 87.6% (600-Labeled,200-Test Data) and 89.3% (700- Labeled,100-Test Data)	Both the metadata and related content features are considered. The results prove it is important to identify fake reviews However, (1) it performs poorly in the short comments, and (2) the model was not compared with baseline models
[48]	YelpCHI ^c	Reported results: Precision: 84.0% Recall: 84.5% F1-score: 81.9%	It achieved good performance compared to traditional algorithms They only used a similar domain dataset
[49]	Extracted from the TripAdvisor	Obtained a maximum accuracy of 87.18% and F1 of 86.70%	The result is lower compared to supervised learning approaches

^a<https://myleott.com/opspam.html>

^b<https://myleott.com/opspam.html>

^c<http://odds.cs.stonybrook.edu/yelpchi-dataset/from>

literature that adopted semi-supervised approaches in spam detection based on different semi-supervised learning approaches.

Lau [50] developed and integrated an unsupervised text mining model into a semantic language model (SLM) for deceptive review detection. They believed that text mining-based methods are more effective for identifying dishonest reviews, regardless of different spammer strategies. The approach was built under the assumption that duplicate reviews (i.e., if the two reviews have similar semantic content) are fake. Spam reviews were identified with the help of cosine similarity, and the reviews with a cosine similarity above a certain threshold are likely to be fake and vice versa. The identification process was also confirmed manually. As tested on Amazon, experimental results showed it is more effective in identifying dishonest

reviews over SVM. However, according to [24], it is unreliable to consider duplicate or near duplicate reviews as spam.

Dong [51] proposed a model to detect fake reviews by mining review topics and sentiments. The model is based on unsupervised learning and latent Dirichlet allocation (LDA). The LDA was employed to extract review topic information. They initially employed the Gibbs sampling algorithm to obtain probabilistic distribution between topics, sentiments, and words and topics. Then, RF and SVM classifiers were fed with the obtained topic and sentiment features. As tested on a dataset from Yelp.com, experimental results showed that the model is better compared to the baselines such as POS, n-grams, and LDA. However, they ignored important features such as reviewer behavior as it is only compared with content-based models.

Based on the nominated topics, Li [52] proposed a three stages model to detect a group of deceptive reviews. In the first stage, they defined equivalent groups with their respective topics; second, to cluster the reviews, they employed the K-means algorithm; third, they used review bustiness and duplicated to cluster reviews. Experimental test results on a real-world dataset collected from JD.com commercial website showed that the solution effectively classifies reviews. However, (1) the proposed method depends only on predefined topics, and (2) according to [24], considering only duplicate or near duplicate content as dishonest is unreliable.

Noekhah [53] proposed a spam review detection model which uses the semantic meaning extracted from reviews text. The model is based on an unsupervised graph. Reported results demonstrated that combining review features improve performance. However, the proposed approach was not compared with recent DL approaches.

Without the labeled dataset, Vidanagama [54] incorporated the related review features such as linguistic, sentiment, and Part-of-Speech (PoS): the domain feature ontology was used in the sentiment feature analysis and extracted review-related features incorporated into the ontology. Then, the spam review detection process was inferencing the ontology. Due to the unavailability of reliable labeled datasets, they used the Mahalanobis distance method to detect outliers that were selected as spam reviews. The proposed approach is convincing. However, its generalizability to other services or domains is questionable since only evaluated on a specific dataset. Additionally, some essential review contexts, such as product/brand popularity and the history of the reviewers, may affect the likelihood of spam reviews. Table 5 summarizes unsupervised learning studies, and datasets used, reported performances, and comments (strengths/weaknesses).

4.2 Neural network-based methods

Recently, NN-based methods have proved to be more powerful in feature extraction, which makes them gain a reputation in various natural language processing (NLP) applications [55, 56]. Unlike basic ML-based approaches, NNs are more effective in fake review detection due to reviews' complex and high-dimensional nature and spammer behaviors [10]. Table 6 briefly describes the basic ML versus NN approaches.

For spam review detection, many of the NN approaches proposed deployed convolutional neural network (CNN), recurrent neural network (RNN), and long short-term memory (LSTM). Originally, CNN was specifically devised for image processing and recently proved to play a significant role in NLP projects in general and fake review classification in specific. RNN is part of DL which identifies sequential characteristics of data to predict the most likely scenario based on data patterns. It processes and saves long sequential information.

Table 5 Unsupervised learning studies, datasets used, reported performances, and comments

Studies	Used datasets	Performances	Comments
[50]	Downloaded from Amazon.com	Reported results: Untruthful reviews: 1-AUC = 0.1346% and Non-reviews 1-AUC = 2.5112%	More effective in identifying dishonest reviews over SVM However, it is unreliable to consider duplicate or near duplicate reviews as spam Mohawesh, Xu, et al., (2021). Furthermore, it is ineffective when the review corpus contains only a single untruthful review
[51]	YelpCHI	Balanced dataset (restaurant): Precision = 82.3%, recall = 85.6%, F1-measure = 83.9%. Unbalanced dataset (restaurant): Precision = 75.25%, recall = 80.6%, F1-measure = 77.8% Balanced dataset (hotel): Precision = 87.2%, recall = 83.0%, F1-measure = 85.0%. Unbalanced dataset (hotel): Precision = 83.6%, recall = 87.3%, F1-measure = 85.4%	Performance depends on the number of the selected topic Its performance is better compared to the baselines such as POS, n-grams, and LDA However, they ignored important features such as reviewer behavior as it is only compared with content-based models
[53]	AMT [39] Amazon.com	Accuracy = 93.0%, and 95.3% on crowdsourced and Ott's datasets, respectively	Results demonstrated that combining review features improve performance However, the proposed approach was not compared with recent DL approaches to show its robustness
[52]	Collected from JD.com	Accuracy = 96.4%	Results demonstrate its effectiveness However, (1) the proposed method depends only on predefined topics, and (2) considering only duplicate or near duplicate content as dishonest is unreliable
[54]	Amazon unlocked mobile phone ^a	Precision = 88.69%, recall = 90.90%, 88.98%, F1 = 89.79%	Model performance may be enhanced by incorporating important review features such as review, reviewer-related, and service/product-related features

^a<https://www.kaggle.com/PromptCloudHQ/amazon-reviews-unlocked-mobile-phones>

Table 6 ML versus NN approaches summary

Basic ML approaches	NN approaches
Trained on predefined features in which they decide depending on pursued data features	There is no need for feature extraction; features are extracted directly from the input dataset. The NN learns and decides intelligently on its own (no or little prior knowledge/interventions required)
Does not require high computational resources and is easy to implement	Require high (sometimes extremely high) computational resources considered <i>black box</i> models since they do not provide a complete understanding of learning (i.e., do not have declarative knowledge to explain the outcomes)
Achieves good results with small datasets and vice versa	More effective on a large amount of data and vice versa

In short, RNN hidden layers depend on word input vectors obtained using a particular word embedding technique, and the last layer can predict the output. However, according to [24], RNNs are ineffective due to vanishing or exploding gradient problems. This subsection sums up the popular existing fake review detection CNN-based models as follows:

4.2.1 CNN-based studies

Li [57] presented a fake review detection model utilizing a neural network to learn the whole review document representation. For training and testing, they used a word vector. The architecture of the proposed model includes (1) a sentence convolutional layer to create a review sentence composition and (2) the document convolutional layer to transform the review sentence vector into a document vector. Experimental evaluation results demonstrated that review document representation performs better on the cross-domain data set than the hand-crafted features. Limitation, the attention weights are computed by a fixed and a hard alignment mode. It may be improved by a more flexible and soft alignment mode. Hence, a memory network-based method might help to overcome the problem.

Wang [58] intended to explore whether the review is linguistically dubious, behaviorally dubious, or both. Therefore, they proposed an attention-based NN to aid in the deceptive review identification process by observing linguistic versus behavioral features. They employed multi-layer perceptron (MLP) and CNN to extract behavioral and linguistic features. Then, an attention mechanism was used to dynamically learn behavioral and linguistic feature weights for every training sample. Experimental results showed the proposed attention mechanism outperforms state-of-the-art models. However, since it is only evaluated on a single-domain dataset, its performance advantage may not generalize.

Considering the cold-start problem, [59] proposed a neural network model that jointly embeds behavioral and textual information for identifying spam reviews. In an unsupervised fashion, it learns to represent the new review with similar textual and behavioral information posted by the new reviewer. They adopted CNN to effectively catch the complex semantic review text information, which is very difficult to catch with traditional techniques such as LIWC and unigram. To encode behavioral information into graph structure (nodes and edges), they used an embedding learning method similar to the TransE [60]. Experimental evaluation results on the dataset extracted on Yelp showed the proposed model achieves better

performance. However, the proposed model was only trained and tested on a single domain. Besides, its effectiveness may depend on the quantity and quality of new reviewer behavioral information.

Zhang [61] proposed the deceptive review identification (DRI) model based on recurrent convolution neural network (RCNN) utilizing the word contexts of the reviews. The basic assumption is that fake and genuine reviews differ in their word representation since spam reviewers lack a real business experience with the product/service described. They suggested the recurrent convolutional vector using its truthful and deceptive contexts and word embedding to represent each review word. The deep neural network (DNN) structure filters fake reviews by integrating the pooling (max-pooling) and ReLU filters. Experimental evaluation results demonstrated the effectiveness of DRI-RCNN in spam filtering as it outperformed state-of-the-art methods. However, it has the following limitations: (1) its feasibility is uncertain in real-world scenarios as training relies on the availability of large amounts of labeled datasets that are not always available, (2) it may not be generalizable well to other languages or domains since reviewer contextual knowledge may vary significantly across them (i.e., languages or domains).

Hajek & Sahut [62] proposed a supervised learning approach that integrates review- and reviewer-centric features for detecting deceptive reviews. First, review sentiment-dependent features extracted, then used CNN for weight matrix fine-tuning and review sentiment classification. The CNN result is combined with behavioral patterns. Finally, to achieve superior classifier performance and effective spam review identification, they recommended that it is vital to (1) jointly integrate behavioral and linguistic aspects and (2) consider the class imbalance problem in the model training process. Limitations of this study: (1) it may not apply to domains other than the hospitality industry, (2) essential factors like reviewer review length and demographics were neglected.

4.2.2 RNN-based studies

Ren & Zhang [63] explored a NN model to learn document representation for spam opinion detection. In particular, CNN is used to learn from sentence representation for a given document, which is integrated using a gated recurrent neural network (GRNN) with an attention mechanism to produce a document vector. Obtained document vector representations were used as features to identify fake reviews. Experimental evaluation results showed proposed approach outperforms several compared state-of-the-art baselines. However, computational complexity analysis of the proposed model is not provided.

Wang [58] utilized the Long Short-term Memory Recurrent Neural Network (LSTM-RNN) framework for review spammers detection. They established a multilayer perceptron with input, hidden, and output layers. First, an input layer receives data as a neuron; after the input layer, LSTM is added, next hidden layer for dimensionality reduction, and finally, an output layer which outputs one neuron. The value of neurons of 1 and 0 means the reviewer predicted as a spammer and a regular user, respectively. Experimental evaluation showed that the LSTM model is more effective in spam review detection than SVM. However, the proposed approach's effectiveness is questionable because (1) not compared with other NN methods, and (2) it focused only on text features; they ignored important review features such as metadata and behavioral features.

Bhuvaneshwari [10] proposed a DL-based framework named "Self-Attention-based Bi-LSTM (ACB)." It learns from document-level representation, the weightage of each word in the review sentence is computed with attention mechanisms, high-level n-gram features extracted using CNN, and then obtained vectors from each review were combined using

Bi-LSTM to identify the fake reviews. Experimental valuation results on YelpZip⁷ datasets demonstrated is more effective than its variants such as CNN-BiLSTM, Spiral Cuckoo Search, Hierarchical Supervised Learning (HSL), CNN, Attention CNN (ACNN), and CNN. However, the paper does not provide the interpretability of the proposed ACB.

The study by [64] aims to classify product reviews as real or fake. To classify reviews, they deployed the CNNLSTM-FABC framework. Important review features extraction (obtained in preprocessing phase) and to minimize processing complications; the representative subset is obtained using the chi-squared method. An experimental dataset which consists of 1500 reviews, was obtained from various including Yelp.com, Expedia, Hotels.com, Priceline, Amazon, and TripAdvisor. Results showed higher performance achieved over its variants: CNN, CNN-LSTM, multilayer perceptron neural network (MLP), and deep neural network (DNN). However, the proposed model was only evaluated on small, balanced, single-domain-based datasets.

Liu [6] proposed a hierarchical attention network for detecting spam reviews where they uses two distinct attention layers to capture only important, complete, and multi-granularity review semantic information. Experimental results on the publicly available datasets demonstrate the proposed approach is superior in spam review detection over its variants. However, we noted the following limitations (1) generalizability is uncertain since the proposed method was evaluated on a limited number of datasets (2) computational complexity details are not provided.

Considering the semantic meaning of the review and emotions hidden in the review [65] proposed two deep NN models (i.e., CNN and deep feed-forward neural network (DFFNN)) that learn from review document-level representation integrating three features: consumer emotion indicators, n-grams, word embeddings features to classify spam reviews. The proposed models, as experimented on four different datasets, namely, doctor, hotel, Amazon, and restaurant, showed that the proposed model's performance is good, regardless of product category and sentiment polarity. However, the lexicon-based emotion indicators may not capture the full range of expressed reviewers' emotions.

Liu [66] presented the Bi-LSTM model for fake review detection. The proposed method learns document-level representation to capture global information by paying attention to the word and sentence order. It is based on feature integration in which review features such as first-person pronoun, word embedding (Glove), and POS were merged and added to the model. Several cross- and mixed-domain experiments were conducted on the dataset released by [67]. Experimental results demonstrated that the proposed model is more effective on the cross- and mixed domains than on its variants. Although the proposed methods require high computational resources, we can notice that the combination of the first-person pronoun, word embedding (Glove), and POS features of review is important in fake review identification.

Alsharif [68] proposed a DL for an e-commerce fake review identification scheme based on linguistic features. The proposed scheme includes (1) Word Embedding Layer: to convert the review text into a numerical vector representation, (2) the Dropout Layer: to prevent the model from over-fitting, (3) LSTM Layer: for processing word embedding vectors, (4) the dense feed-forward layer: to connect all available neurons in a network, and (5) Sigmoid Activation Function: to performs for review classification (i.e., spam or non-spam opinion). The reported performance was 98% in terms of F1-score and accuracy, which is better than existing approaches. However, we noted the two main weaknesses: (1) generalization is uncertain as they tested only on a single and monolingual dataset, and (2) the proposed method relies on extracted review linguistic features. Relying only on linguistic features may

⁷ <https://myleott.com/opspace.html>.

not be effective since spam reviews are sometimes written to sound/mimic genuine ones. Table 7 summarizes Neural network-based studies and datasets used, reported performances, and comments (strengths/weaknesses).

4.3 Other fake review detection approaches

Li [52] proposed and used a review group concept algorithm to segment and group reviewers' reviews for fake reviews classification (i.e., fake or not). Additionally, they built the reviewer group collusion model based on the reviewer's collusion relationship. Evaluations showed significant improvement against baselines. However, the proposed method was not compared to some of the latest techniques used for group spam detection. Besides, the experimental dataset used may not be representative of other domains.

Manaskasemsak [3] proposed two graphs-based fake review(er)s detection algorithms named behavioral graph partitioning (BeGP) and BeGPX. The BeGPX is an extension of BeGP introduced purposely to enhance detection performance. The graph construction was built based on four aspects of reviews, namely, emotional indicators, word embedding feature representation, review, and reviewer information. They used the deep neural network to learn the lexicon-based indicators of emotion and word embeddings feature vector representation. Then, the review and reviewers' statistics are integrated to complete nodes of the graph. The presented evaluation result outperforms the compared baselines. However, we noted the following: (1) the proposed method relies on the assumption that spammers always exhibit similar behavior, which is not always the case, (2) datasets used are limited to Yelp.com, the performance advantage may not be feasible for other domains, and (3) it requires labeled data for training, reliable labeled data are not always available.

You [1] proposed an unsupervised model named an aspect rating local outlier factor (AR-LOF) for deceptive review detection that involves obtaining specific aspect ratings from free text. It utilizes tensor factorization, sentiment lexicon, and aspect ranking algorithms to get more fine-grained aspect-based spam review detection. The sentiment lexicon, in particular, is used to obtain the specific rating of aspects. Results demonstrated the effectiveness of the proposed model in fake review detections. However, the authors did not provide or compare their approach with state-of-the-art baselines. Also, it may not be suitable in complex scenarios such as review datasets based on the cross or mixed domain, multilingual, or language mix (code-mix).

In their part, [43] addressed feature pruning and imbalanced data issues, proposing an ensemble model to filter online spam reviews. The framework consists of four interdependent step techniques, namely; (1) data resampling and grid search: to address the problem of data imbalance, (2) feature pruning: to drop insignificant features, (3) parameters optimization: to obtain suitable parameter values, and (4) classifier ensemble: to integrate the optimized base classifiers through applying stacking and majority voting strategies. Evaluation results showed achieved the best results. However, it only outperforms some existing techniques, and the proposed ablation study for feature pruning is time-consuming, especially when so many features are included.

Several existing methods which use complete review text analyses face the computation time complexity problem. Bathla [73] addressed this problem by proposing the review aspects extraction and respective sentiments in identifying spam reviews. First, extracted aspects of review and respective sentiments are inputted into CNN for replication, and CNN's output is then loaded into LSTM as input for classification. They obtained good results over their counterparts as tested on two datasets. Limitations of this study are: (1) although it is a recent

Table 7 summarizes neural network-based studies, datasets used, reported performances, and comments

Studies	Used datasets	Performances	Comments
[69]	Doctor, restaurant, and hotel datasets released by [70]	Accuracy = 80.1%, F1 value = 86.1%	Experimental results demonstrate that the proposed approach is very robust in cross-domain spam detection. SWNN is very efficient compared to other NN-based methods Limitation, the attention weights are computed by a fixed and a hard alignment mode. It may be improved by a more flexible and soft alignment mode. So, a memory network-based method might help to overcome the problem
[58]	Extracted from the Yelp website	On the hotel domain: Precision = 88.1%, recall = 89.7%, F1 = 88.9%, and accuracy = 88.8% On the restaurant domain: Precision = 89.4%, recall = 93.0%, F1 = 91.2%, and accuracy = 91.0%	Performance shows the importance of behavioral and linguistic features in attention mechanism review classification However, to identify deceptive reviews, the model mainly focused on linguistic features which are not enough compared to behavioral features
[59]	Extracted on Yelp.com	On the hotel domain (RE + RRE + PRE): Precision = 63.6%, recall = 71.2%, F1 = 67.2%, and accuracy = 65.2% On the restaurant domain (RE + RRE + PRE): Precision = 59.0%, recall = 78.8%, F1 = 67.5%, and accuracy = 62.0%	The cold-start problem affecting effective fake review detection was analyzed. In an unsupervised fashion, it learns to represent the new review with similar textual and behavioral information posted by the new reviewer In dealing with the cold-start problem, CNN is more effective compared to the traditional model
[61]	Spam, and deception datasets from [35], and [70], respectively	The best performance accuracy is at training percent of 0.9 = 88.2% with a standard deviation of 0.1442	Achieved good performances However, they ignored behavioral features which might boost classification performances

Table 7 (continued)

Studies	Used datasets	Performances	Comments
[62]	YelpZip ^a	AUC = 0.916, F1 = 0.830	The study neglected the aspects level review analysis and emotions hidden in the review, which are essential indicators of fake content
[63]	Three domain datasets (hotel, restaurant, and doctor)[71]	CNN: accuracy = 75.9%, Macro-F1 = 77.4% RNN: accuracy = 63.2%, Macro-F1 = 64.8% GRNN: accuracy = 80.1%, Macro-F1 = 80.7% Bi-directional: accuracy = GRNN 83.6%, Macro-F1 = 83.4% Bi-directional GRNN (attention): accuracy = 84.1%, Macro-F1 = 83.9%	Results show Bi-directional GRNN (attention) is superior However, the study neglected the aspects level review analysis and emotions hidden in the review, which are essential indicators of fake content
[72]	Real-world reviews (Mobile01.com) in Taiwan,	Accuracy = 0.894, F measure = 0.796	Experimental results demonstrated that LSTM is superior to conventional SVM, so they recommended DL to detect fake reviews effectively However, they did not compare with NN methods
[10]	YelpZip ^b	Accuracy = 86%, precision = 90%, recall = 84%, F1 = 87%	Obtained good results However, it neglected the relationship between overall star ratings and aspect ratings
[64]	Personalized dataset extracted from Amazon, Yelp, and TripAdvisor	Accuracy 98%, recall = 93%, F1 = 95%	No statistical tests were performed to support the dominance of the proposed model
[65]	hotel ^c , restaurant, doctor, and Amazon ^d datasets	On amazon dataset: accuracy = 82%, F1 = 82%, AUC = 89% On hotel: accuracy = 89%, F1 = 95%, AUC = 89% Restaurant: accuracy = 88%, F1 = 88%, AUC = 95% Doctor: accuracy = 86%, F1 = 89%, AUC = 94%	Good reported performance regardless of dataset size, type (category), and sentiment polarity However, the review sentence weights, the product, and reviewer-based features were not fully explored

Table 7 (continued)

Studies	Used datasets	Performances	Comments
[66]	Gold standard dataset released by [70]	Cross domain restaurant: accuracy = 0.813, precision = 0.778, recall = 0.875, F1 = 0.824 Cross domain doctor: accuracy = 0.668, precision = 0.631, recall = 0.805, F1 = 0.708 Mixed domain: accuracy = 0.852, precision = 0.845, recall = 0.909, F1 = 0.876	They reported good performance However, it requires higher computational resources
[68]	Yelp's product reviews	Achieved 0.98 in terms of F1 and accuracy	Reported results are very high However, no statistical tests were performed to support the dominance of the proposed model
[6]	Used a public dataset released by [70]	F1 score in the mixed domain improved up to 0.89 (F1 score of 0.93 and 0.86 in the Doctor, and Hotel domains, respectively) F1 score in the cross-domain improved to 84.7%	It demonstrated its superiority by utilizing multi-granularity and comprehensive review information to detect fake reviews. However, (1) no special attention was paid to the discrete features of the review, and (2) the credibility of the supervised learning approaches is questionable due unavailability of reliable labeled datasets (see Sect. 5)

^a<http://odds.cs.stonybrook.edu/yelpzip-dataset/>

^b<http://odds.cs.stonybrook.edu/yelpzip-dataset/>

^c<http://myleott.com/op-spam.html>

^d<https://www.kaggle.com/lievgarciamazon-reviews>

study, authors still relied on an old and non-contextualized form of embedding, and (2) it tested on monolingual datasets.

Some recent findings recommended multi-feature combinations for effective spam review identification. Inspired by this recommendation [74], it focused on determining the best feature combination for effective deceptive review detection among topic distributions, sentiment scores, bag of words, and cluster distributions. The findings of this research pointed out that integrating review text features and behavior-related features plays a vital role in review classifications. Although the study focuses on discovering the best feature integration, it only evaluated limited or commonly used review features. It is expected to be tested on a different angle of review features such as source reliability and reviewer reputation properties.

Zhang [75] proposed ImDetector to identify dishonest reviewers taking into account the data imbalance problem based on Kullback–Leibler (KL) divergence and weighted LDA.

The weighted LDA was used to extract latent topics described in the review, while KL divergence was adopted to measure the similarity among reviewers biased on the fraudulent minority. They also adopted the kNN for classification. Experiments were conducted on the Yelp.com dataset to demonstrate its effectiveness over its counterparts. This study has the following limitations (1) the review text contents were not explored, (2) ignored group reviewer behaviors (only considered individual reviewer behaviors).

4.4 Non-English fake review detection techniques

As introduced earlier, we have noted significant efforts geared primarily toward English-based review datasets compared to other languages. Few or no fake review detection studies exist, even on some popular non-English, and low-resource languages spoken by a large population. Besides, based on literature surveys, we can confidently state that Chinese is the second most popular language explored in fake review detection. This subsection describes the following non-English-based studies on fake review detection. Chinese: [71, 76–82], Malay [83], Arabic: [84, 85], Korean [86], Roman Urdu [87], Spanish[88], and Portuguese [88].

The most compelling studies on fake review detection based on Chinese include but are not limited to [71] proposed a fake review detection approach in Chinese. The study reported the two supervised learning classes, namely fake and unknown. They assumed it is more important to treat an unknown class as unlabeled since it may contain a number of fake reviews. It calls for the learning model from positive and unlabeled (PU-learning). Experiment evaluation results demonstrated that PU learning significantly outperforms supervised learning and detects many potentially fake reviews hidden in the unlabeled set. Weng [89] proposed a pre-trained language model for detecting deceptive Chinese reviews based on BERT. This study achieved good performance results as reported: 92% precision, 91% F1-score, and recall. In similar fake review detection studies on Chinese e-commerce platforms, [82] proposed three spam indicator groups: merchant, individual, and group indicators. They employed an unsupervised matrix iteration algorithm to calculate fake degree values in each indicator group. This study achieved good performance compared to the baselines. However, they did not consider how each indicator contributes to falsities; they only took averages of related fake indicators directly. Xu [90] proposed KNN- and graph-based classification approaches to address collusive spammer indicators in Chinese reviews. Based on reviewers' group-level relational information, KNN- considers the pairwise similarity between them, and then the most k similar reviewers are selected for voting. Based on reviewers' pairwise transaction correlations, the graph-based classification approach classifies a set of reviewers jointly.

Antonio [79] developed a multilingual model predicting online review ratings (English, Spanish, and Portuguese) to identify fraudulent reviews. The study highlighted that significant deviations between predicted and actual ratings are good signals of potential fraud, suggesting discrepancies in quantitative and qualitative components. Reviews exceeding a set threshold are then considered potential fraudulent postings.

Hazim [83] proposed a supervised boosting approach, namely generalized boosted regression model (GBM) and extreme gradient boost (XGBoost), to detect spam reviews. The proposed model was evaluated on the multilingual dataset (i.e., English and Malay languages). Reported comparative analysis showed that the XGBoost is more effective in the English-based dataset (achieved 87.43% accuracy), while the GBM Gaussian is more suitable for the Malay dataset (achieved 86.13% accuracy). However, they relied on a custom-made

Malay dataset since there are no publicly available Malay datasets. For generalization, there is a need for evaluation of the standard gold dataset.

Ma [86] utilizes data from a South Korean movie review site (movie.naver.com) to investigate the influence of review manipulation on box office revenue. The study focuses on dynamic review manipulation behavior, examining review distribution patterns and the impact of promotional reviews during the early stages of a movie's release. It emphasizes the susceptibility of unverified reviews to manipulation and discusses how consumers perceive a higher quantity and positivity of unverified reviews as an indication of manipulation, triggering psychological reactance.

Hayat [87] explored different DL methods based on the Roman Urdu language. First, they constructed a new review corpus (i.e., Roman Urdu). The developed corpus contains 5150 reviews (3398/1752 train/test instances). Second, several experimental tests were conducted using DL architectures: simple RNN, LSTM, GRU, Bi-LSTM, and Bi-GRU. The reported evaluation showed stacked LSTM model achieved great results ($F1 = 0.88$ and $ACC-ROC = 0.943$). However, results may not be applicable to other languages as they tested only Roman Urdu.

Hammad [84] introduced first-time spam review detection in Arabic language reviews by merging data and text mining techniques into a single mining classification approach. They used sampling techniques (under-sample and over-sample) to overcome the class imbalance problem. Experimental results based on ML, such as SVM, NB) and KNN, proved its effectiveness. The reported best-case result achieved a 99.59% F-measure in identifying Arabic spam comments. However, the approach's effectiveness may largely depend on the quantity and quality of the available data.

In their part, [85] contributes to Arabic spam review detection. They presented their proposed approach, which mainly emphasizes constructing and evaluating an ensemble methods-based rule-based classifier integration with ML techniques utilizing content-based features that depend on negation handling and n-gram features. Reported evaluation results on two variable-size datasets indicate the effectiveness of the proposed ensemble method as it achieved an accuracy of 99.98% and 95.25% in the first and second datasets, respectively. However, it may not be applicable to other languages as it only focuses on Arabic opinions.

4.5 Features used for fake review detection

Throughout the literature, the following are the main features used to aid in training deceptive review detection models. We grouped them into two: (1) textual review features that include but are not limited to semantic details, lexicons, grammar, and metadata features, and (2) behavioral features, which are the abnormal or normal statistical records of reviewers. Particularly in spam review detection, multi-aspect feature integration frameworks outperform single-aspect-based frameworks [91–93]. In their study, Mukherjee et al. [94] compared the effectiveness of using behavioral versus textual features and concluded that utilizing textual features is less effective compared to behavioral features of review. In addition, relying only on the text aspect is inadequate, as the wording difference between spam and non-spam reviews is insignificant [59, 95, 96]. Furthermore, due to the sparsity problems, [55] and [97] explicitly highlighted that it is difficult to spot the semantic meaning by only using review text features. Tables 8 and 9 describe the common textual and behavioral features used in the existing literature.

Table 8 Textual features of reviews

Features	Descriptions
Semantic features	They build and present the underlying meaning of review texts. According to [98], utilizing semantic features of review is better than PoS and LIWC However, they cannot capture the underlying relationship that exists between words and documents
Part of Speech (PoS) tagging	PoS frequency in the review text can be used as a PoS feature Based on the review context, the PoS feature extraction process essentially involves tagging (1) the review word features with PoS and (2) the related and adjacent words [99] However, BoW features are still more effective in detecting spam reviews regardless of good performances achieved with POS features in cross-domain [24, 35, 70]
LIWC	It is a popular feature in NLP projects to analyze different aspects of linguistic features However, they are less effective in deceptive review detection than n-gram features [35, 70]
Metadata information	They are specific review aspects or features, which include review/reviewer ID, review length, feedback text, ratings, usefulness, brand name, review date, and time However, metadata features are not always available in many datasets
Stylometric (linguistic style)	Represent syntactic features of the review covering the types of words and characters used by the reviewer. In addition, it covers various writing styles of reviewers, such as the number of punctuations, the number of upper cases that appear in a review, and the average word length
Bag-of-words (BoW)	They were widely used in various NLP projects and fake review detection in specific They represent review text as a single or continuous number of words Also known as n-grams (unigram, bigram, and trigram) However, they cannot capture the review's semantic meaning

5 Popular datasets for fake review detection and future directions

5.1 Fake review detection datasets and collection techniques

Datasets play an important role in the training and testing of fake review detection frameworks. Table 10 summarizes the popular and public datasets used in fake review detections; it consists of dataset names, general descriptions (dataset size and key features), and where used (studies) that achieved good performance. Generally, datasets collection techniques are: (1) manual labeling or deploying AMT to generate spam reviews in which spammers called Turkers are invited to voluntarily write fake reviews [4, 46, 69, 70, 93, 112–114], (2) Filtering tool (algorithms) such as Yelp spam filter used to label reviews as either genuine or fake [70, 102, 115], filtering algorithms basically use the user's behavioral and lexical features to learn and classify the reviews, (3) Fakespot,⁸ this is an online tool used for spotting reliability of customer reviews [116], and (4) Crawling (web-scraper process) and apply a rule to classify reviews on e-commerce sites such as Amazon, TripAdvisor, Yelp, Dianping, Taobao (Tmall),

⁸ <https://myleott.com/opspam.html>.

Table 9 Behavioral features of reviews

Features	Descriptions
Maximum content similarity and repeated reviews	The similarity in the review's comment describing different services/products can indicate a spammer [50, 93, 100, 101]. Review contents similarity can be captured with the help of cosine similarity Any repeated reviews posted by the same or different reviewers may indicate fraud However, according to [18], this feature needs to be considered carefully, as it is possible for the same reviewer to appear posting the same review repetitively due to internet of system connectivity issues
Review length	It is a very useful reviewer-centric feature Spam reviews are usually short and vice versa, sometimes without details on specific products or services. Hence, this characteristic can help to identify review spammers [69, 92, 102–104]
Percentage of positive/negative review	Observation shows that most spammer's reviews are usually positive (highly) or negative (smallest) rated. Hence a high percentage of positive/negative ratings on a particular product/service might be a good indicator of dishonest reviews [33, 102]
Overall rating deviation	Spammers intentionally attempt to, respectively, rate high or low to promote or demote a service/product [103] Most dishonest rating scores usually vary considerably from an average rating. Since spam ratings are usually high, it can help to detect spam reviews [57, 103, 105]
Aspects ratings	Analysis of review aspect-level rating targets at discrete extractions topical and associated opinions expressed by reviewers. Aspects rating shows customer or reviewer satisfaction with the specific aspect [106]. For example, Table 11 shows three reviews from the same hotel, but each reviewer emphasizes a different aspect regardless of having the same overall star rating value According to [107], reviews datasets either contain inconsistent overall rating values or are completely wrong regarding the respective service/product described. Moreover, You [1] highlighted that despite the overall star rating values similarity in some reviews, spam and non-spam reviews differ in their aspect rating values. It is obvious since spammers usually lack real business experience compared to actual reviewers

Table 9 (continued)

Features	Descriptions
Reviewers emotions	Emotion words in the review can determine the reviewer's positive, neutral, and negative feelings [108, 109] Fake reviews typically have more emotionally charged words and fewer factual numbers [110, 111]. Since the prime spammer's intention is to either promote or demote the product/service, they usually try to make their review look authentic (real), but, difficult to hide their emotions

Expedia, Priceline, ResellerRatings, Airbnb, Appstore, Playstore. For example, using user behavior and content features [96] crawled through review datasets obtained from Yelp.com to label them based on a different predefined set of rules (rule-based method), that the repeated reviews or text similarity are labeled spam. However, the availability and reliability of the collected gold-standard dataset are still questionable due to the following reasons;

- It is impractical to have Turkers with the same psychological state of mind when writing fake reviews as real deceivers on a commercial website who have real business interests to promote or to demote1). Similarly, Turkers may not have sufficient domain knowledge or experience to write convincing fake reviews [4, 93, 103, 117].
- Due to the multidimensional context and dynamic nature of review information, manual annotation (labeling) is very expensive in terms of time and manpower. Thus, it may result in inaccurately labeled kind of datasets. In addition, human beings cannot effectively distinguish fake from benign reviews [24, 33, 35].
- Issues with filtering tools (algorithms): since these tools rely on behavioral and lexical features, they might not be reliable in some circumstances. For instance, a VPN and proxy server can easily hide spammer devices (IP addresses), making locating the reviewer's geographical location challenging. Besides, due to the sparsity problem, it isn't easy to spot the semantic meaning of a review by only using linguistic or text features [55, 97]. Additionally, relying only on text features is uncertain since spammers are very smart today in wording their reviews to sound similar to benign ones, especially on commercial websites [59, 95, 96].
- The reliability of datasets obtained by employing rule-based methods might sometimes be questionable. For instance, [18] labeled the datasets deploying several rules that the review is fake if: (1) different reviewers comment on the same service or product, (2) the same reviewers comment on the same service or product, or (3) different reviewers comment on different service or products. Considering these rules to label review is uncertain because sometimes, due to misconfiguration (network connection problems), the same honest customer (reviewers) may appear, posting several likely reviews on the same or different product or service simultaneously. Hence, it is inappropriate to generalize all similar (duplicate or near duplicate) reviews as fake based on these rules.

Furthermore, we observed: (1) a lack of datasets covering multilingual and/or language code-mix aspects. For example, in countries with official duo languages, it is common to find code-mix language-based reviews, (2) a lack of enough datasets with multidimensional review feature information. Most existing labeled datasets, including gold-standard [35, 70],

Table 10 Summary of popular and public datasets used in fake reviews detection

Datasets	Descriptions	Studies used the dataset
TripAdvisor ^a [45]	Total reviews: 800 positive reviews, where; 400 are genuine and 400 deceptive Features: Text only	[46, 55, 65, 73, 85, 104, 111, 118–127]
TripAdvisor ^b [39]	Total reviews: 800 negative reviews where; 400 are genuine, and 400 deceptive Features: Text only	[46, 55, 104, 118–120, 123, 125, 128]
YelpCHI ^c [93]	Total reviews: 67,395 restaurant and hotel reviews, where 58,476 are genuine and 8,919 deceptive Features: review text (plain text), overall star ratings values, user information, and timestamp	[41, 43, 48, 93, 95, 115, 125, 129–131]
YelpNYC ^d [102]	Total reviews: 359,052 restaurant reviews where; 322,167 are genuine, and 36,885 deceptive Features: review text (plain text), overall star ratings values, user information, and timestamp	[3, 41, 58, 95, 129–135]
YelpZIP ^e [102]	Total reviews: 608,598 restaurant reviews where; 528,132 are genuine, and 80,466 deceptive Features: review text (plain text), overall star ratings values, user information, and timestamp	[3, 10, 41, 62, 95, 123, 129–133, 135–139]
Amazon ₂ ^f [18]	Total reviews: 5.83 million product reviews Features: Product and reviewers IDs, overall ratings, review date, review title, review body, helpfulness/feedback	[18, 132, 136, 140, 141]
Amazon ₃ ^g	Total reviews: 21,000 reviews, where 10,500 are genuine and 10,500 deceptive, respectively Features: overall rating, product ID, product category, verified purchase (yes or no)	[65, 74]

Table 10 (continued)

Datasets	Descriptions	Studies used the dataset
Restaurant, and Doctor [70]	Total reviews (restaurant): 400 reviews, where 200 and 200 are genuine and deceptive, respectively Total reviews (doctor): 556, where 200 and 356 are genuine and deceptive, respectively	[6, 55, 65, 119]
Amazon ₁ ^h [142]	Total reviews: 142.8 million products reviews Features: overall ratings, review text, review helpfulness votes, and product metadata features such as product category, descriptions, brand, price, and image features	[143, 144]
HARD (Hotel Arabic-Reviews Dataset) ⁱ [145]	Total reviews: 94,052 opinion reviews in Arabic, covering 1858 hotels and sourced from Booking.com Features: review text in the Arabic language, rating (on a scale of 1 to 10 stars), and hotel/reviewer attributes Not labeled	[85, 85, 145, 146, 148]
RU-FRDC ^j [87]	Total reviews: 5,150 (3398 for training, and 1752 for testing) Features: Review text only Labeled	[87, 149, 150]
Dianping ^k	Consists of reviews from 500 restaurants in Shanghai, China Features: Review text, product and reviewer information Review consists of filtered (fake) reviews and unfiltered (unlabeled) reviews	[71, 98, 151]

^a<https://myleott.com/opspam.html>

^b<https://myleott.com/opspam.html>

^c<http://odds.cs.stonybrook.edu/yelpchi-dataset/>

^d<http://odds.cs.stonybrook.edu/yelpnyc-dataset/>

^e<http://odds.cs.stonybrook.edu/yelpnyc-dataset/>

^fThe dataset can be obtained freely upon request in <http://liu.cs.uic.edu/download/data/>

^g<https://www.kaggle.com/lievgarciamazon-reviews>

^h<https://nijianmo.github.io/amazon/index.html>

ⁱ<https://github.com/elngara/HARD-Arabic-Dataset>

^j<https://github.com/MuhammadFarhatUllah/Roman-Urdu-Fake-Review-Detection>

^k<http://liu.cs.uic.edu/download/dianping> (This dataset is available upon request, please send email to lhymvp@gmail.com)

lack important review features (contain only text), such as the metadata information required for model training.

6 Conclusion

In this work, we first introduce the motivation behind fake review detection on e-commerce platforms. Then, we summarize the research designs and review structures, presenting systematic steps for accomplishing the study effectively, including planning, conducting the survey, and reporting stages. Through our research design, we collect several state-of-the-art studies and provide a comprehensive analysis detailing the strengths and weaknesses, features used for fake review detection, and popular datasets. Publications for this study were retrieved from notable journals and conferences spanning from 2007 to 2022. Despite the remarkable significant effort to enhance the online business's reliability, the following issue still needs more attention.

6.1 Availability of unquestionable gold-standard datasets

Most popular labeled gold-standard datasets were collected in similar methodologies. For example, AMT effectively generated and labeled fake review datasets by hiring anonymous online Turkers (freelancers). However, the credibility of the collected datasets remains questionable because of the following facts: (1) they do not reflect the dynamics of real fake reviews on a real commercial website, (2) assuming all Turkers have the same psychological state of mind when preparing these pseudo-fake reviews is unrealistic similar to real deceivers who have a real commercial motive to either demote or promote, and (3) most benchmark datasets were collected over a decade ago, they may not align with recent technological advancements, such as the invention of large language models like ChatGPT, where spammers can easily craft reviews to mimic authentic ones. In addition, as introduced earlier, filtering algorithms rely on behavioral and lexical features. They might not be reliable since behavior features can easily be avoided. For example, spammers can hide from geographical location inspection using a VPN or proxy server. Besides, rule-based methods might sometimes be inappropriate, as explained earlier. Hence, the generalizability of models resulting from training on these questionable datasets (pseudo-fake reviews) is unrealistic. In discourse matter, unsupervised frameworks have attempted to overcome this matter, yet their performance results are still lower compared to supervised approaches. We recommend the adoption of a hybrid approach for generating reliable datasets, as hybrid approaches have shown promising results in covering a wide range of positive aspects of the review.

6.2 Language code-switching and multilingual review spam detection

Different language representation challenges exist, i.e., to investigate the efficiency of the proposed solution on review presented in multiple languages. Existing spam review detection approaches are based on monolingual, such as Chinese, English, and Arabic.

Table 11 Sample reviews with similar overall ratings but a different emphasis on described aspects of the same hotel

Overall rating	Review text
3	"Overpriced for what you get. The staff was nice but our room was terrible."
3	"The food was very good, but it took over half an hour to be seated, ... and the service was terrible. The room was very noisy and cold wind blew in from a curtain next to our table. Desserts were very good, but because of poor service, I'm not sure we'll ever go back"
3	"It is new and clean, and the rate of 99 euros per night was very reasonable. It is difficult to find on foot. Their WIFI service is not available in rooms."

6.3 Deviation of overall rating scores with review sentiment values

Dishonest ratings usually vary considerably from the average rating [103]. Several studies addressed the fake reviews problem by considering only rating deviation [57, 103, 105]. From the same perspective, future researchers should assess the impact of deviation of overall rating scores with review sentiment values.

6.4 Review aspects' ratings

The reviews' ratings are not extensively explored in spam review detections. According to [1], genuine and fake reviews differ in the described aspects in association with the respective rating values regardless of how similar they are in overall rating values. For example, Table 11 contains reviews with the same overall star rating but different emphases on service provided in hotel aspects such as price, room, WiFi, service, and location. Reviewers always describe aspects of a service or product to convince others or to show how bad or good the service or product is. However, since spammers usually lack real business experience, it is apparent aspects described by spammers could be different from genuine ones, regardless of the overall rating similarity. In this regard, future works must focus on the aspect rating dimension. Most existing approaches consider overall positive or negative emotions or the overall star rating of a review. Considering only coarse overall ratings is ineffective, as it fails to represent all possible aspects of the review dimensions.

6.5 The optimal set of features for integration

Some existing works have suggested and outperformed recent state-of-the-art baselines through multi-feature integration covering review- and reviewer-centric aspect features. Still, they have not explicitly highlighted the optimal set of features for integration, as model performance results can be slightly affected by not considering some review features [152, 153]. In addition, some existing studies consider review burstiness to be a good indicator of spam [52, 57, 92, 107]. However, they failed to assess the modest rate of distribution statistically, the ratio of reviewers' negative/positive emotions in the review, and the rating distribution.

Furthermore, several recent works attempted a combination of multi-aspect features (reviewer- and review-centric features) with promising performance. Still, they ignored the correlation between them and the lack of a unified framework that learns from all important review features such as coarse-grained (overall rating), fine-grained (aspects ratings),

and emotions to supplement the problem of sentiment, and overall rating inconsistency still deserves special attention. Additionally, although heterogeneous graphs are currently gaining interest in various real-world applications, most studies on deceptive review detection still focus on homogeneous graphs, irrespective of the diverse nature of reviews in real-world scenarios. Also, review text (comment) and overall ratings are mutually correlated. However, existing review-based approaches only consider reviewer-review, product-review, or reviewer-product interactions, ignoring multiple interactions between reviewer and product features, such as reviewer-review-product and reviewer-rating-product interactions.

In summary, we hope that this work will be helpful for researchers in the field of fake review detection by providing critical insights into existing studies, gold-standard datasets, and future research directions. We believe that addressing the challenges highlighted in this work will lead to more accurate and reliable fake review detection methods, which can help ensure fair e-business competition and promote sustainable online practices.

Author contributions Ramadhani Duma was involved in the conceptualization, formal analysis, investigation, resources, and writing—original draft. Zhendong Niu and Ally Nyamawe contributed to the methodology, resources, supervision, validation, editing and review, and project administration. Jude Tchaye-Kondi, Abdulganiyu Yusuf, Nuru Jingili, and Augustino Deve performed the editing and review

Funding This work was supported by the National Natural Science Foundation of China under Grant 62272048.

Declarations

Conflict of interest All the authors declare no conflict of interest.

Ethical approval For this kind of study, there is nothing that requires ethical approval.

Informed consent No formal consent is required for this kind of study.

References

1. You L, Peng Q, Xiong Z, He D, Qiu M, Zhang X (2020) Integrating aspect analysis and local outlier factor for intelligent review spam detection. *Futur Gener Comput Syst* 102:163–172. <https://doi.org/10.1016/j.future.2019.07.044>
2. Khan ZY, et al (2021) A deep hybrid model for recommendation by jointly leveraging ratings, reviews and metadata information. *Eng Appl Artif Intell* 97:104066
3. Manaskasemsak B, Tantisuwankul J, Rungsawang A (2021) Fake review and reviewer detection through behavioral graph partitioning integrating deep neural network. *Neural Comput Appl* 1:1169–1182. <https://doi.org/10.1007/s00521-021-05948-1>
4. Vidanagama DU, Silva TP, Karunananda AS (2020) Deceptive consumer review detection: a survey. *Artif Intell Rev* 53(2):1323–1352. <https://doi.org/10.1007/s10462-019-09697-5>
5. Duma RA, et al (2024) DHMFRD–TER: a deep hybrid model for fake review detection incorporating review texts, emotions, and ratings. *Multimedia Tools Appl* 83(2):4533–4549
6. Liu Y, Wang L, Shi T, Li J (2022) Detection of spam reviews through a hierarchical attention architecture with N-gram CNN and Bi-LSTM. *Inf Syst* 103:101865. <https://doi.org/10.1016/j.is.2021.101865>
7. Ren J, Yeoh W, Shan Ee M, Popović A (2018) Online consumer reviews and sales: examining the chicken-egg relationships. *J Assoc Inf Sci Technol* 69(3):449–460. <https://doi.org/10.1002/asi.23967>
8. Yu S, Ren J, Li S, Naseriparsa M, Xia F (2022) Graph learning for fake review detection. *Front Artif Intell* 5:1–14. <https://doi.org/10.3389/frai.2022.922589>
9. Ellson A (2021) A third of TripAdvisor reviews are fake' as cheats buy five stars | The Times, Accessed December, 20 2021," 2018. <https://www.thetimes.co.uk/article/hotel-and-caf-cheats-are-caught-trying-to-buy-tripadvisor-stars-027fbwcw8>. Accessed 20 Dec 2021.

10. Bhuvaneshwari P, Rao AN, Robinson YH (2021) Spam review detection using self attention based CNN and bi-directional LSTM. *Multimed Tools Appl* 80(12):18107–18124. <https://doi.org/10.1007/s11042-021-10602-y>
11. Luca M (2016) Reviews, reputation, and revenue : the case of Yelp.com.com (March 15, 2016). Harvard Bus. Sch. NOM Unit Work. Pap., pp 12–016.
12. Crawford M, Khoshgoftaar TM, Prusa JD, Richter AN, Al Najada H (2015) Survey of review spam detection using machine learning techniques. *J Big Data*. <https://doi.org/10.1186/s40537-015-0029-9>
13. Palmer A (2023) People are using A.I. chatbots to write Amazon reviews. CNBC, 2023. <https://www.cnbc.com/2023/04/25/amazon-reviews-are-being-written-by-ai-chatbots.html>. Accessed 16 Feb 2024
14. Goldrich MJ (2024) The rise of AI-generated fake reviews: what you need to know. VIVANDER ADVISORS. <https://vivanderadvisors.com/the-rise-of-ai-generated-fake-reviews-what-you-need-to-know/>. Accessed 16 Feb 2024
15. Wu J (2024) AI-generated fake reviews: a new challenge for online trust. Medium. <https://bootcamp.uxdesign.cc/ai-generated-fake-reviews-a-new-challenge-for-online-trust-87e6ed825a80>. Accessed 17 Feb 2024
16. Gambetti A, Han Q (2023) Combat AI with AI: counteract machine-generated fake restaurant reviews on social media. vol. 1, no. 1. Association for Computing Machinery, 2023. [Online]. Available: <http://arxiv.org/abs/2302.07731>
17. Krishna K, Song Y, Karpinska M, Wieting J, Iyyer M (2023) Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. no. NeurIPS, pp. 1–32. [Online]. Available: <http://arxiv.org/abs/2303.13408>
18. Jindal N, Liu B (2008) Opinion spam and analysis. In: WSDM'08 - Proc. 2008 Int. Conf. Web Search Data Min., pp 219–229. <https://doi.org/10.1145/1341531.1341560>
19. Mewada A, Dewang RK (2021) Research on false review detection Methods: A state-of-the-art review. *J King Saud Univ Comput Inf Sci* 18:100. <https://doi.org/10.1016/j.jksuci.2021.07.021>
20. Ren Y (2014) Positive unlabeled learning for deceptive reviews detection. pp 488–498
21. Heydari A, Tavakoli MA, Salim N, Heydari Z (2015) Detection of review spam: a survey. *Expert Syst Appl* 42(7):3634–3642. <https://doi.org/10.1016/j.eswa.2014.12.029>
22. Patel NA, Patel R (2018) A survey on fake review detection using machine learning techniques. In: 2018 4th Int. Conf. Comput. Commun. Autom. ICCCA 2018, pp 1–6. <https://doi.org/10.1109/CCAA.2018.8777594>.
23. Ren Y, Ji D (2019) Learning to detect deceptive opinion spam: a survey. *IEEE Access* 7:42934–42945. <https://doi.org/10.1109/ACCESS.2019.2908495>
24. Mohawesh R et al (2021) Fake reviews detection: a survey. *IEEE Access* 9:65771–65802. <https://doi.org/10.1109/ACCESS.2021.3075573>
25. Maurya SK, Singh D, Maurya AK (2022) Deceptive opinion spam detection approaches: a literature survey. *Appl Intell* 53:2189–2234
26. He L, Wang X, Chen H, Xu G (2022) Online spam review detection : a survey of literature. *Hum Centric Intell Syst* 2:14–30
27. Kitchenham B et al (2010) Systematic literature reviews in software engineering-a tertiary study. *Inf Softw Technol* 52(8):792–805. <https://doi.org/10.1016/j.infsof.2010.03.006>
28. Page MJ et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372:n71. <https://doi.org/10.1136/bmj.n71>
29. Crawford M, Khoshgoftaar TM, Prusa JD, Richter AN, Al Najada H (2015) Survey of review spam detection using machine learning techniques. *J Big Data* 18:10. <https://doi.org/10.1186/s40537-015-0029-9>
30. Wahyuni ED, Djunaidy A (2016) Fake review detection from a product review using modified method of iterative computation framework. *MATEC Web Conf*. <https://doi.org/10.1051/mateconf/20165803003>
31. Banerjee S, Kim J (2015) Using supervised learning to classify authentic and fake. In: Proc 9th Int Conf Ubiquitous Inf Manag Commun ACM, New, pp 1–7
32. Shojaei S, Murad MAA, Bin Azman A, Sharef NM, Nadali S (2013) Detecting deceptive reviews using lexical and syntactic features. In: 2013 13th International Conference on Intelligent Systems Design and Applications, pp 53–58. <https://doi.org/10.1109/ISDA.2013.6920707>
33. Li F, Huang M, Yang Y, Zhu X (2011) Learning to identify review spam. In: *IJCAI Int Jt Conf Artif Intell*, pp 2488–2493. <https://doi.org/10.5591/978-1-57735-516-8/IJCAI11-414>
34. Hernández-castañeda Á, Calvo H, Gelbukh A, Flores JGG (2017) Cross-domain deception detection using support vector networks. *Soft Comput* 21(3):585–595. <https://doi.org/10.1007/s00500-016-2409-2>
35. Ott M, Choi Y, Cardie C, Hancock JT (2011) Finding deceptive opinion spam by any stretch of the imagination. *arXiv Prepr. arXiv1107.4557*

36. Interno M, Sicurezza P, Dipartimento S, Poesio M (2014) Identifying fake Amazon reviews as learning from crowds. pp 279–287.
37. Mihalcea R, Science C, Science C (2014) Cross-cultural deception detection. pp 440–445
38. Mani S, Kumari S, Jain A, Kumar P (2018) Spam review detection using ensemble machine learning. Springer International Publishing. <https://doi.org/10.1007/978-3-319-96133-0>
39. Ott M, Cardie C, Hancock JT (2013) Negative deceptive opinion spam. In: Proceedings of the 2013 Conference of the North American chapter of the Association for Computational Linguistics: Human Language Technologies, pp 497–501
40. Sánchez-junquera J, Villaseñor-pineda L, Montes-y-gómez M, Rosso P, Stamatatos E (2020) Masking domain-specific information for cross-domain deception detection. *Pattern Recognit Lett* 135:122–130. <https://doi.org/10.1016/j.patrec.2020.04.020>
41. Mohawesh R, Tran S, Ollington R, Xu S (2021) Analysis of concept drift in fake reviews detection. *Expert Syst Appl* 169:2021
42. Shan G, Zhou L (2021) From conflicts and confusion to doubts : examining review inconsistency for fake review detection. *Decis Support Syst* 144:113513
43. Yao J, Zheng Y, Jiang H (2021) An ensemble model for fake online review detection based on data resampling, feature pruning, and parameter optimization. *IEEE Access* 9:16914–16927. <https://doi.org/10.1109/ACCESS.2021.3051174>
44. Tufail H, Ashraf MU, Alsubhi K, Aljahdali HM (2022) The effect of fake reviews on e-commerce during and after Covid-19 pandemic: SKL-based fake reviews detection. *IEEE Access* 10:25555–25564. <https://doi.org/10.1109/ACCESS.2022.3152806>
45. Ott M, Cardie C, Hancock J (2012) Estimating the prevalence of deception in online review communities. In: WWW'12—Proc 21st Annu Conf World Wide Web, pp 201–210. <https://doi.org/10.1145/2187836.2187864>
46. Rout JK, Dalmia A, Choo KKR, Bakshi S, Jena SK (2017) Revisiting semi-supervised learning for online deceptive review detection. *IEEE Access* 5:1319–1327. <https://doi.org/10.1109/ACCESS.2017.2655032>
47. Deng H, Zhao L, Luo N, Liu Y, Guo G, Wang X (2017) Semi-supervised learning based fake review detection. <https://doi.org/10.1109/ISPA/IUCC.2017.00195>
48. Wang J, Kan H, Meng F, Mu Q, Shi G, Xiao X (2020) Fake review detection based on multiple feature fusion and rolling collaborative training. *IEEE Access* 8:182625–182639. <https://doi.org/10.1109/access.2020.3028588>
49. Jing-Yu C, Ya-Jun W (2022) Semi-supervised fake reviews detection based on AspmGAN. *J Artif Intell Capsul Netw* 4(1):17–36. <https://doi.org/10.36548/jaicn.2022.1.002>
50. Lau RYKK, Liao SY, Chi-Wai Kwok R, Xu K, Xia Y, Li Y (2011) Text mining and probabilistic language modeling for online review spam detection. *ACM Trans Manag Inf Syst* 2(4):1–30. <https://doi.org/10.1145/2070710.2070716>
51. Dong LY et al (2018) An unsupervised topic-sentiment joint probabilistic model for detecting deceptive reviews. *Expert Syst Appl* 114:210–223. <https://doi.org/10.1016/j.eswa.2018.07.005>
52. Li J, Lv P, Xiao W, Yang L, Zhang P (2021) Exploring groups of opinion spam using sentiment analysis guided by nominated topics. *Expert Syst Appl* 171:114585. <https://doi.org/10.1016/j.eswa.2021.114585>
53. Noekhah S, Binti Salim N, Zakaria NH (2020) Opinion spam detection: Using multi-iterative graph-based model. *Inf Process Manag* 57(1):102140. <https://doi.org/10.1016/j.ipm.2019.102140>
54. Vidanagama DU, Silva ATP, Karunananda AS (2022) Ontology based sentiment analysis for fake review detection. *Expert Syst Appl* 206:117869. <https://doi.org/10.1016/j.eswa.2022.117869>
55. Ren Y, Ji D (2017) Neural networks for deceptive opinion spam detection: an empirical study. *Inf Sci (Ny)* 385–386:213–224. <https://doi.org/10.1016/j.ins.2017.01.015>
56. Lauriola I, Lavelli A, Aiolfi F (2022) An introduction to deep learning in natural language processing: models, techniques, and tools. *Neurocomputing* 470:443–456. <https://doi.org/10.1016/j.neucom.2021.05.103>
57. Li H et al. (2017) Bimodal distribution and co-bursting in review spam detection. In: 26th Int. World Wide Web Conf. WWW 2017, pp 1063–1072. <https://doi.org/10.1145/3038912.3052582>
58. Wang X, Liu K, Zhao J (2018) Detecting deceptive review spam via attention-based neural networks. In: *Lecture Notes Computer Science* (including subseries *Lecture Notes in Artificial Intelligence* and *Lecture Notes in Bioinformatics*), vol 10619 LNAI, pp 866–876. https://doi.org/10.1007/978-3-319-73618-1_76
59. Wang X, Liu K, Zhao J (2017) Handling cold-start problem in review spam detection by jointly embedding texts and behaviors. In: *ACL 2017 - 55th Annu. Meet. Assoc. Comput. Linguist. Proc. Conf. (Long Pap., vol. 1)*, pp. 366–376, 2017, doi: <https://doi.org/10.18653/v1/P17-1034>.
60. Bordes A, Usunier N, Garcia-dur A, Weston J, Yakhnenko O (2013) Translating embeddings for modeling multi-relational data. *Adv Neural Inf Process*, pp 2787–2795

61. Zhang W, Du Y, Yoshida T, Wang Q (2018) DRI-RCNN: an approach to deceptive review identification using recurrent convolutional neural network. *Inf Process Manag* 54(4):576–592. <https://doi.org/10.1016/j.ipm.2018.03.007>
62. Hajek P, Sahut JM (2021) Mining behavioural and sentiment-dependent linguistic patterns from restaurant reviews for fake review detection. *Technol Forecast Soc Chang* 177:121532. <https://doi.org/10.1016/j.techfore.2022.121532>
63. Ren Y, Zhang Y (2016) Deceptive opinion spam detection using neural network. In: *proceedings of COLING 2016, the 26th international conference on computational linguistics: technical papers*. pp 140–150, Osaka, Japan. COLING 2016 Organ. Committee., pp 140–150
64. Jacob MS, Rajendran PS (2021) Fuzzy artificial bee colony-based CNN-LSTM and semantic feature for fake product review classification. *Concurr Comput Pract Exp* 18:10. <https://doi.org/10.1002/cpe.6539>
65. Hajek P, Barushka A, Munk M (2020) Fake consumer review detection using deep neural networks integrating word embeddings and emotion mining. *Neural Comput Appl* 32(23):17259–17274. <https://doi.org/10.1007/s00521-020-04757-2>
66. Liu W, Jing W, Li Y (2020) Incorporating feature representation into BiLSTM for deceptive review detection. *Computing* 102(3):701–715. <https://doi.org/10.1007/s00607-019-00763-y>
67. Li J, Ott M, Cardie C, Hovy E (2014) Hausarzt-Rechnung-22-08-11.Pdf. pp 1566–1576
68. Alsharif N (2011) Fake opinion detection in an e-commerce business based on a long-short memory algorithm. *Soft Comput* 5:2022. <https://doi.org/10.1007/s00500-022-06806-5>
69. Li L, Qin B, Ren W, Liu T (2017) Document representation and feature combination for deceptive spam review detection. *Neurocomputing* 254:33–41. <https://doi.org/10.1016/j.neucom.2016.10.080>
70. Li J, Ott M, Cardie C, Hovy E (2014) Towards a general rule for identifying deceptive opinion spam. In: *Proceedings of 52nd annual meeting of the association for computational linguistics: system demonstrations*, vol 1, pp 1566–1576. <https://doi.org/10.3115/v1/p14-1147>
71. Li H, Chen Z, Liu B, Wei X, Shao J (2014) Spotting fake reviews via collective positive-unlabeled learning. In: *IEEE international conference on data mining (ICDM)*, vol 2015, pp 899–904. <https://doi.org/10.1109/ICDM.2014.47>
72. Wang C, Chen C, Day M-Y, Liou J-W (2018) Detecting spamming reviews using long short-term memory recurrent neural network framework. In: *Proceeding of the 2nd international conference on E-commerce, E-business, and E-government*, pp 16–20
73. Bathla G, Singh P, Singh RK, Cambria E, Tiwari R (2022) Intelligent fake reviews detection based on aspect extraction and analysis using deep learning. *Neural Comput Appl* 8:20213. <https://doi.org/10.1007/s00521-022-07531-8>
74. Birim ŞÖ, Kazancoglu I, Kumar Mangla S, Kahraman A, Kumar S, Kazancoglu Y (2021) Detecting fake reviews through topic modelling. *J Bus Res* 149:884–900. <https://doi.org/10.1016/j.jbusres.2022.05.081>
75. Zhang W, Xie R, Wang Q, Yang Y, Li J (2022) A novel approach for fraudulent reviewer detection based on weighted topic modelling and nearest neighbors with asymmetric Kullback-Leibler divergence. *Decis Support Syst* 157:113765. <https://doi.org/10.1016/j.dss.2022.113765>
76. Liu L, Zhao X, Wang H, Song W, Du C (2016) Research on identification method of anonymous fake reviews in E-commerce. *TELKOMNIKA Telecommun Comput Electron Control* 14(4):1510. <https://doi.org/10.12928/telkomnika.v14i4.3654>
77. Zhang L, Wu Z, Cao J (2017) Detecting spammer groups from product reviews: a partially supervised learning model. *IEEE Access* 6:2559–2568. <https://doi.org/10.1109/ACCESS.2017.2784370>
78. Zhang L, He G, Cao J, Zhu H, Xu B (2018) Spotting review spammer groups: a cosine pattern and network based method. *Concurr Comput Pract Exp* 30(20):1–15. <https://doi.org/10.1002/cpe.4686>
79. Gao X, Li S, Zhu Y, Nan Y, Jian Z, Tang H (2019) Identification of deceptive reviews by sentimental analysis and characteristics of reviewers. *J Eng Sci Technol Rev* 12(1):196–202. <https://doi.org/10.25103/jestr.121.23>
80. Xu Y, Yang Y, Han J, Wang E, Ming J, Xiong H (2019) Slandorous user detection with modified recurrent neural networks in recommender system. *Inf Sci (NY)* 505:265–281. <https://doi.org/10.1016/j.ins.2019.07.081>
81. Liu Y, Pang B, Wang X (2019) Opinion spam detection by incorporating multimodal embedded representation into a probabilistic review graph. *Neurocomputing* 366:276–283. <https://doi.org/10.1016/j.neucom.2019.08.013>
82. Yu S, Su J, Luo D (2019) Improving BERT-based text classification with auxiliary sentence and domain knowledge. *IEEE Access* 7:176600–176612. <https://doi.org/10.1109/ACCESS.2019.2953990>
83. Hazim M, Anuar NB, Ab Razak MF, Abdullah NA (2018) Detecting opinion spams through supervised boosting approach. *PLoS ONE* 13(6):1–23. <https://doi.org/10.1371/journal.pone.0198884>

84. Hammad AA, El-Halees A (2015) An approach for detecting spam in Arabic opinion reviews. *Int Arab J Inf Technol* 12(1):10–16
85. Saeed RMK, Rady S, Gharib TF (2019) An ensemble approach for spam detection in Arabic opinion texts. *J. King Saud Univ Comput Inf Sci* 34(1):1407–1416. <https://doi.org/10.1016/j.jksuci.2019.10.002>
86. Ma H, Kim JM, Lee E (2019) Analyzing dynamic review manipulation and its impact on movie box office revenue. *Electron Commer Res Appl*. <https://doi.org/10.1016/j.elerap.2019.100840>
87. Hayat U, Saeed A, Vardag MHK, Ullah MF, Iqbal N (2022) Roman Urdu Fake reviews detection using stacked LSTM architecture. *SN Comput Sci*. <https://doi.org/10.1007/s42979-022-01385-6>
88. Almeida D, Almeida D (2018) Hotel online reviews: creating a multi-source aggregated index. *Int J Contemp Hosp Manag* 30(12):3574–3591
89. Weng CH, Lin KC, Ying JC (2022) Detection of Chinese deceptive reviews based on pre-trained language model. *Appl Sci* 12(7):3338. <https://doi.org/10.3390/app12073338>
90. Xu C, Zhang J, Chang K, Long C (2013) Uncovering collusive spammers in Chinese review websites. *Int Conf Inf Knowl Manag Proc*. <https://doi.org/10.1145/2505515.2505700>
91. Jindal N, Liu B (2007) Review spam detection. In: 16th International World Wide Web conference, WWW2007, pp 1189–1190. <https://doi.org/10.1145/1242572.1242759>
92. Fei G, Mukherjee A, Liu B, Hsu M, Castellanos M, Ghosh R (2013) Exploiting burstiness in reviews for review spammer detection. In: Proceedings of the seventh international conference on Weblogs and social media, ICWSM 2013, vol 7, no 1, pp 175–184
93. Mukherjee A, Venkataraman V, Liu B, Glance N (2013) What yelp fake review filter might be doing?. In: Proceedings of the 7th International conference on Weblogs and social media, ICWSM 2013, pp 409–418
94. Mukherjee A, Liu B, Glance N (2012) Spotting fake reviewer groups in consumer reviews. In: WWW'12: Proceedings of the 21st annual conference on World Wide Web, pp 191–200. <https://doi.org/10.1145/2187836.2187863>
95. Budhi GS, Chiong R, Wang Z (2021) Resampling imbalanced data to detect fake reviews using machine learning classifiers and textual-based features. *Multimed Tools Appl* 80(9):13079–13097. <https://doi.org/10.1007/s11042-020-10299-5>
96. Barbado R, Araque O, Iglesias CA (2019) A framework for fake review detection in online consumer electronics retailers. *Inf Process Manag* 56(4):1234–1244. <https://doi.org/10.1016/j.ipm.2019.03.002>
97. Zhang Y, Wallace B (2015) A sensitivity analysis of (and practitioners' guide to) convolutional neural networks for sentence classification, 2015. [Online]. Available: <http://arxiv.org/abs/1510.03820>
98. Li H, Chen Z, Mukherjee A, Liu B, Shao J (2015) Analyzing and detecting opinion spam on a large-scale dataset via temporal and spatial patterns. *Proc 9th Int Conf Web Soc Media* 9:634–637
99. Hussain N, Mirza HT, Rasool G, Hussain I, Kaleem M (2019) Spam review detection techniques: a systematic literature review. *Appl Sci* 9(5):987. <https://doi.org/10.3390/app9050987>
100. Serrano-guerrero J, Olivás JA, Romero FP, Herrera-viedma E (2015) Sentiment analysis: a review and comparative analysis of web services. *Inf Sci (NY)* 311:18–38. <https://doi.org/10.1016/j.ins.2015.03.040>
101. Akram AU, Khan HU, Iqbal S, Iqbal T, Munir EU, Cantt W (2018) Finding rotten eggs : a review spam detection model using diverse feature sets. vol 12, no 10, pp 5120–5142
102. Rayana S, Akoglu L (2015) Collective opinion spam detection: bridging review networks and metadata. In: Proceedings of the ACM SIGKDD international conference on knowledge discovery and data mining, vol 2015-August, pp 985–994. <https://doi.org/10.1145/2783258.2783370>
103. Mukherjee A et al. (2013) Spotting opinion spammers using behavioral footprints. In: Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, vol. Part F1288, pp 632–640. <https://doi.org/10.1145/2487575.2487580>
104. Mohawesh R, Xu S, Springer M, Al-Hawawreh M, Maqsood S (2021) Fake or genuine? Contextualised text representation for fake review detection. pp 137–148. <https://doi.org/10.5121/csit.2021.112311>
105. Lim EP et al. (2010) Detecting product review spammers using rating behaviors. In: Proc. 19th {ACM} Conf. Inf. Knowl. Manag. {CIKM} 2010, Toronto, Ontario, Canada, Oct. 26–30, 2010 Manag. {CIKM} 2010, Toronto, Ontario, Canada, Oct. 26–30, 2010, pp 939–948. <https://doi.org/10.1145/1871437.1871557>
106. Wang H, Lu Y, Zhai C (2010) Latent aspect rating analysis on review text data: a rating regression approach. In: Proceedings of the ACM SIGKDD international conference on knowledge discovery and data mining, pp 783–792. <https://doi.org/10.1145/1835804.1835903>
107. Li J, Fu Y, Liu D, Xu R (2020) Improving fake product detection with aspect-based sentiment analysis. In: International conference on cognitive computing, pp 39–49. https://doi.org/10.1007/978-3-030-59585-2_4
108. Denecke K (2008) Using SentiWordNet for multilingual sentiment analysis. In: Proceedings international conference on data engineering., pp 507–512. <https://doi.org/10.1109/ICDEW.2008.4498370>

109. Malik MSI, Hussain A (2017) Helpfulness of product reviews as a function of discrete positive and negative emotions. *Comput Hum Behav* 73(2010):290–302. <https://doi.org/10.1016/j.chb.2017.03.053>
110. Li Y, Feng X, Zhang S (2016) Detecting fake reviews utilizing semantic and emotion model. In: 2016 3rd international conference on information science and control engineering (ICISCE), pp 317–320. <https://doi.org/10.1109/ICISCE.2016.77>
111. Saini M, Sharan A (2017) Identifying deceptive opinion spam using aspect-based emotions and human behavior modeling. *Int J Hybrid Inf Technol* 10(1):447–456. <https://doi.org/10.14257/ijhit.2017.10.1.36>
112. Plotkina D, Munzel A, Pallud J (2020) Illusions of truth—experimental insights into human and algorithmic detections of fake online reviews. *J Bus Res* 109:511–523. <https://doi.org/10.1016/j.jbusres.2018.12.009>
113. Ong T, Mannino M, Gregg D (2014) Linguistic characteristics of shill reviews. *Electron Commer Res Appl* 13(2):69–78. <https://doi.org/10.1016/j.elerap.2013.10.002>
114. Huang YK, Yang WI, Lin TMY, Shih TY (2012) Judgment criteria for the authenticity of internet book reviews. *Libr Inf Sci Res* 34(2):150–156. <https://doi.org/10.1016/j.lisr.2011.11.006>
115. Arjun M, Vivek V, Bing L, Natalie G (2013) Fake review detection: classification and analysis of real and pseudo reviews. *Tech Rep* 80(2):159–169
116. Wu Y, Ngai EWT, Wu P, Wu C (2020) Fake online reviews: literature review, synthesis, and directions for future research. *Decis Support Syst* 132:113280. <https://doi.org/10.1016/j.dss.2020.113280>
117. Rout JK, Singh S, Jena SK, Bakshi S (2017) Deceptive review detection using labeled and unlabeled data. *Multimed Tools Appl* 76(3):3187–3211. <https://doi.org/10.1007/s11042-016-3819-y>
118. Hernández Fusilier D, Montes-y-Gómez M, Rosso P, Guzmán Cabrera R (2015) Detecting positive and negative deceptive opinions using PU-learning. *Inf Process Manag* 51(4):433–443. <https://doi.org/10.1016/j.ipm.2014.11.001>
119. Cagnina LC, Rosso P (2017) Detecting deceptive opinions: intra and cross-domain classification using an efficient representation. *Int J Uncertain Fuzziness Knowl Based Syst* 25:151–174. <https://doi.org/10.1142/S0218488517400165>
120. Martinez-Torres MR, Toral SL (2019) A machine learning approach for the identification of the deceptive reviews in the hospitality sector using unique attributes and sentiment orientation. *Tour Manag* 75(June):393–403. <https://doi.org/10.1016/j.tourman.2019.06.003>
121. Rajamohana SP, Umamaheswari K, Abirami B (2017) Performance analysis of iBPSO and BFPA based feature selection techniques for improving classification accuracy in review spam detection. *Appl Math Inf Sci* 11(4):1149–1153. <https://doi.org/10.18576/amis/110422>
122. Zhang W, Bu C, Yoshida T, Zhang S (2016) CoFea: a novel approach to spam review identification based on entropy and co-training. *Entropy* 18(12):429. <https://doi.org/10.3390/e18120429>
123. Ansari G, Ahmad T, Doja MN (2018) Spam review classification using ensemble of global and local feature selectors. *Cybern Inf Technol* 18(4):29–42. <https://doi.org/10.2478/cait-2018-0046>
124. Pandey AC, Rajpoot DS (2019) Spam review detection using spiral cuckoo search clustering method. *Evol Intell* 12(2):147–164. <https://doi.org/10.1007/s12065-019-00204-x>
125. Cardoso EF, Silva RM, Almeida TA (2018) Towards automatic filtering of fake reviews. *Neurocomputing* 309:106–116. <https://doi.org/10.1016/j.neucom.2018.04.074>
126. Kleinberg B, Mozes M, Arntz A, Verschuere B (2018) Using named entities for computer-automated verbal deception detection. *J Forensic Sci* 63(3):714–723. <https://doi.org/10.1111/1556-4029.13645>
127. Zhang W, Bu C, Yoshida T, Zhang S (2016) CoSpa: a co-training approach for spam review identification with support vector machine. *Information* 7(1):12. <https://doi.org/10.3390/info7010012>
128. Kim J, Kang J, Shin S, Myaeng SH (2021) Can you distinguish truthful from fake reviews? User analysis and assistance tool for fake review detection. In: *Proceedings of the First Workshop on Bridging Human–Computer Interaction and Natural Language Processing. HCINLP 2021 - Proc. 1st Work.*, pp 53–59
129. Yilmaz CM, Durahim AO (2018) SPR2EP: a semi-supervised spam review detection framework. In: *Proceedings of the 2018 IEEE/ACM international conference on advance social networks analysis mining, ASONAM 2018*, pp 306–313. <https://doi.org/10.1109/ASONAM.2018.8508314>
130. A. Melleng, A. J. Loughrey, and P. Deepak, “Sentiment and emotion based text representation for fake reviews detection,” *Int. Conf. Recent Adv. Nat. Lang. Process. RANLP*, vol. 2019–Septe, no. 2019, pp. 750–757, 2019, doi: https://doi.org/10.26615/978-954-452-056-4_087.
131. Cao C, Li S, Yu S, Chen Z (2021) Fake reviewer group detection in online review systems. In: *IEEE international conference on data mining workshops (ICDMW)*, vol. 2021, pp. 935–942. <https://doi.org/10.1109/ICDMW53433.2021.00122>
132. Wang Z, Gu S, Zhao X, Xu X (2018) Graph-based review spammer group detection. *Knowl Inf Syst* 55(3):571–597. <https://doi.org/10.1007/s10115-017-1068-7>

133. Liu M, Shang Y, Yue Q, Zhou J (2021) Detecting fake reviews using multidimensional representations with fine-grained aspects plan. *IEEE Access* 9:3765–3773. <https://doi.org/10.1109/ACCESS.2020.3047947>
134. Yuan C, Zhou W, Ma Q, Lv S, Han J, Hu S, Learning review representations from user and product level information for spam detection. In: *Proceedings under IEEE international conference on data mining (ICDM)* | IEEE Conference, vol 2019, pp 1444–1449. <https://doi.org/10.1109/ICDM.2019.00188>.
135. Shaalan Y, Zhang X, Chan J, Salehi M (2021) Detecting singleton spams in reviews via learning deep anomalous temporal aspect-sentiment patterns. vol 35, no 2. Springer US. <https://doi.org/10.1007/s10618-020-00725-5>
136. Shehnpoor S, Salehi M, Farahbakhsh R, Crespi N (2017) NetSpam: a network-based spam detection framework for reviews in online social media. *IEEE Trans Inf Forensics Secur* 8(6):748–752
137. Kumar N, Venugopal D, Qiu L, Kumar S (2019) Detecting anomalous online reviewers: an unsupervised approach using mixture models. *J Manag Inf Syst* 36(4):1313–1346. <https://doi.org/10.1080/07421222.2019.1661089>
138. Kumar J (2018) Fake review detection using behavioral and contextual features fake review detection using behavioral and contextual features
139. Xu C, Zhang J (2017) Collusive opinion fraud detection in online reviews: a probabilistic modeling approach. *ACM Trans Web* 11(4):1–28. <https://doi.org/10.1145/3098859>
140. Savage D, Zhang X, Yu X, Chou P, Wang Q (2015) Detection of opinion spam based on anomalous rating deviation. *Expert Syst Appl* 42(22):8650–8657. <https://doi.org/10.1016/j.eswa.2015.07.019>
141. Sun C, Du Q, Tian G (2016) Exploiting product related review features for fake review detection. vol 2016, no 1
142. Ni J, Li J, McAuley J (2019) Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In: *2019 Conference on Empirical Methods in Natural Language Processing and 9th International Joint Conference on Natural Language Processing*, pp 188–197. <https://doi.org/10.18653/v1/d19-1018>.
143. Kauffmann E, Peral J, Gil D, Ferrández A, Sellers R, Mora H (2019) A framework for big data analytics in commercial social networks: a case study on sentiment analysis and fake review detection for marketing decision-making. *Ind Mark Manag*. <https://doi.org/10.1016/j.indmarman.2019.08.003>
144. Asghar MZ, Ullah A, Ahmad S, Khan A (2020) Opinion spam detection framework using hybrid classification scheme. *Soft Comput* 24(5):3475–3498. <https://doi.org/10.1007/s00500-019-04107-y>
145. Elnagar A, Khalifa YS, Einea A (2018) Hotel arabic-reviews dataset construction for sentiment analysis applications. *Stud Comput Intell* 740:35–52. https://doi.org/10.1007/978-3-319-67056-0_3
146. Al-Mutawa RF, Al-Aama AY (2023) User opinion prediction for Arabic hotel reviews using Lexicons and artificial intelligence techniques. *Appl Sci* 13(10):5985. <https://doi.org/10.3390/app13105985>
147. Saeed RMK, Rady S, Gharib TF (2022) An ensemble approach for spam detection in Arabic opinion texts. *J King Saud Univ Comput Inf Sci* 34(1):1407–1416. <https://doi.org/10.1016/j.jksuci.2019.10.002>
148. Berrimi M, Oussalah M, Moussaoui A, Saidi M (2023) Attention mechanism architecture for Arabic sentiment analysis. *ACM Trans Asian Low-Resour Lang Inf Process*. <https://doi.org/10.1145/3578265>
149. Ullah MF, Saeed A (2023) BERT model for Roman Urdu fake review identification. pp 0–15
150. Attri V (2023) Enhancement of fake reviews classification using deep learning hybrid models. *J Surv Fish Sci* 10:3254–3272
151. Li H et al. Modeling review spam using temporal patterns and co-bursting behaviors. 2016. [Online]. Available: <http://arxiv.org/abs/1611.06625>
152. Luo N, Deng H, Zhao L, Liu Y, Wang X, Tan Z (2017) Multi-aspect feature based neural network model in detecting fake reviews. In: *2017 4th International Conference on Information Science and Control Engineering*, pp 475–479. <https://doi.org/10.1109/ICISCE.2017.106>
153. Duma R, Niu Z, Nyamawe A, Tchaye-kondi J, Yusuf A (2023) A Deep hybrid model for fake review detection by jointly leveraging review text, overall ratings and aspects ratings. *Soft Comput* 27:1–24. <https://doi.org/10.1007/s00500-023-07897-4>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Ramadhani Ally Duma received his BSc in Information Technology in 2010 from Tumaini University, Iringa University College (IUCo.), and his MSc in Computer Science in 2015 from the University of Dar-es-Salaam, Dar-es-Salaam, Tanzania. He is currently pursuing a Ph.D. in Computer Science at the Beijing Institute of Technology (BIT). He works as an Assistant Lecturer at the Information Systems and Technology (IST), College of Information and Virtual Education (CIVE), University of Dodoma (UDOM). His research interests include Machine Learning, fake reviews detection, and e-learning techniques.



Zhendong Niu received his Ph.D degree in Computer Science from Beijing Institute of Technology in 1995. He is currently a Professor and Deputy Dean of the School of Computer Science and Technology, Beijing Institute of Technology, China. He was a postdoctoral researcher at the University of Pittsburgh from 1996 to 1998, and a researcher/adjunct faculty member at Carnegie Mellon University from 1999 to 2004, and a joint research Professor in the Information School at University of Pittsburgh from 2006. His research areas include digital libraries, e-learning techniques, information retrieval and recommender systems. He serves as an editorial board member for international journal of learning technology. He has published more than 100 papers in journals and international conferences in his field.



Ally S. Nyamawe received the B.Sc. degree in computer science from the University of Dar es Salaam, Tanzania, in 2008, and the M.Sc. degree in computer science from the University of Dodoma, Tanzania, in 2011. He is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Beijing Institute of Technology, China. He is currently a Lecturer with the Department of Computer Science, University of Dodoma. His research interests include software refactoring, requirements engineering, requirements traceability, and computer programming.



Jude Tchaye-Kondi received the BS degree from the Department of Computer Science, Catholic University of West Africa. He joins in 2017 Beijing Institute of Technology, China as a graduate student and is currently pursuing his Ph.D. His research interest includes parallel and distributed computing, edge intelligence, machine learning.



Nurul Jingili, PhD computer science 2020, is a seasoned researcher in the field of computer science, with a focus on Persuasive Technologies and Artificial Intelligence. She has extensive experience as a project manager and coordinator for initiatives such as the GreenGaming and WeSteam projects from 2022 to 2024. Her current research endeavours delve into various aspects of Artificial Intelligence, Persuasive Technologies, Gamified Interactive Systems, Virtual Reality, Web Technologies, Mobile Computing, and Human-Computer Interaction.



Abdulganiyu Abdu Yusuf received his BSc in Computer Science from Kano University of Science and Technology, Nigeria, in 2008; MSc in Artificial Intelligence from Universiti Sultan Zainal Abidin, Malaysia, in 2015. He is currently a doctoral researcher in the School of Computer Science and Technology, Beijing Institute of Technology, China. He work as a Senior Computer Analyst at National Biotechnology Development Agency. His research interests include computer vision, image processing and deep learning. He received the best graduating student award in both BSc and MSc level and won excellent PhD student award twice.



Augustino Faustino Deve earned his Bachelor of Science in Information, Communication and Technology from The Open University of Tanzania, Tanzania, in 2011. Currently, he is pursuing a Ph.D. in the school of Computer Science at Beijing Institute of Technology. His research focuses on sign language detection and recognition, deep neural network exploration, object detection, and instance segmentation.

Authors and Affiliations

**Ramadhani Ally Duma^{1,2} · Zhendong Niu¹ · Ally S. Nyamawe² ·
Jude Tchaye-Kondi¹ · Nuru Jingili³ · Abdulganiyu Abdu Yusuf¹ ·
Augustino Faustino Deve^{1,4}**

✉ Zhendong Niu
zniu@bit.edu.cn

Ramadhani Ally Duma
radsiffi@gmail.com

Ally S. Nyamawe
ally.nyamawe@uodom.ac.tz

Jude Tchaye-Kondi
tchaye59@gmail.com

Nuru Jingili
nurujingili@gmail.com

Abdulganiyu Abdu Yusuf
abdulg720@gmail.com

Augustino Faustino Deve
tngeszwa@gmail.com

¹ School of Computer Science and Technology, Beijing Institute of Technology, Beijing 100081, People's Republic of China

² College of Informatics and Virtual Education, The University of Dodoma, Dodoma, Tanzania

³ Luleå University of Technology, Skellefteå, Sweden

⁴ Nelson Mandela African Institution of Science and Technology, Arusha, Tanzania